DNA Artifact Classification
Machine Learning Model Comparison Report

Report Generated: December 22, 2025

PROJECT OVERVIEW

This report presents a comprehensive comparison of machine learning models for DNA artifact classification in Low Template DNA (LTDNA) analysis.

CLASSIFICATION TASK

- Target Classes: Allele, Back_Stutter, Forward_Stutter, Pull_Up
- Models Evaluated: 9 models
- Resampling Strategies: Original, SMOTE, RandomOverSampler

MODELS ANALYZED

- TabPFN (Tabular Prior-Fitted Networks)
- LightGBM
- XGBoost
- Extra Trees Classifier
- Random Forest
- Logistic Regression
- Support Vector Machine
- Multi-Layer Perceptron
- Gaussian Naive Bayes

EVALUATION METRICS

- Accuracy & Balanced Accuracy
- F1-Score (Macro)
- Mean Squared Error (MSE) & Mean Absolute Error (MAE)
- Log Loss
- Per-Class Precision, Recall, F1-Score
- Confusion Matrices
- ROC & Precision-Recall Curves
- Feature Importance (tree-based models)
- SHAP Values (select models)

Overall Model Performance Comparison

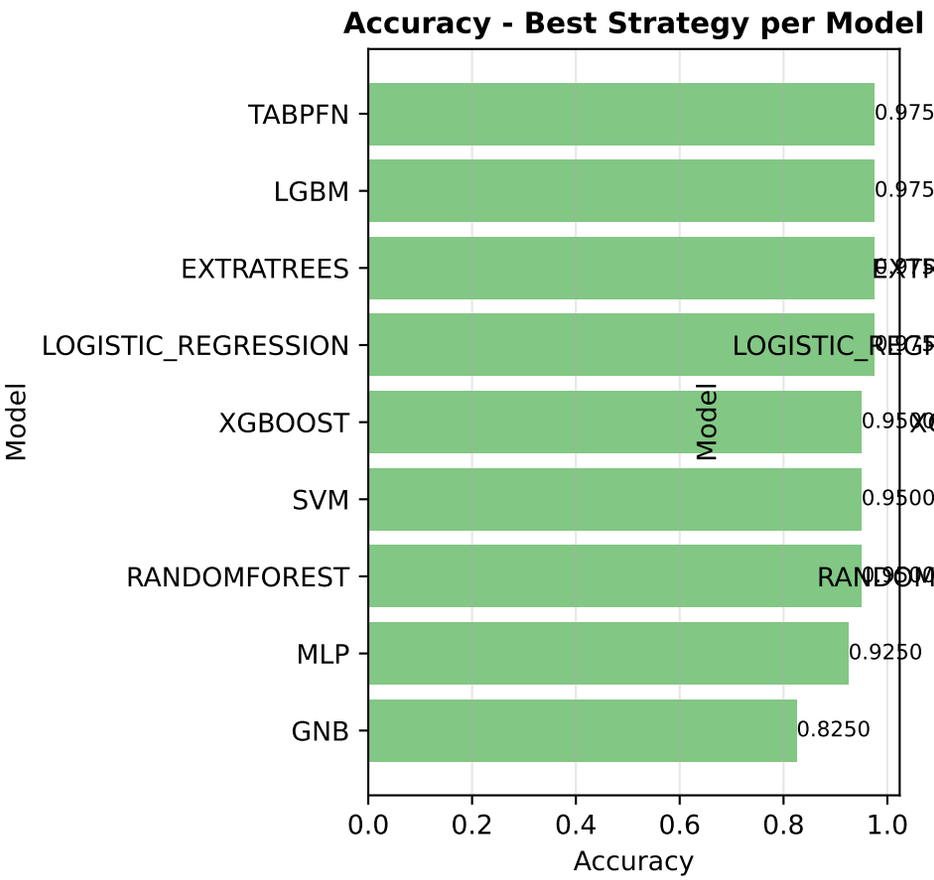
Top 10 Model-Strategy Combinations (by Balanced Accuracy)

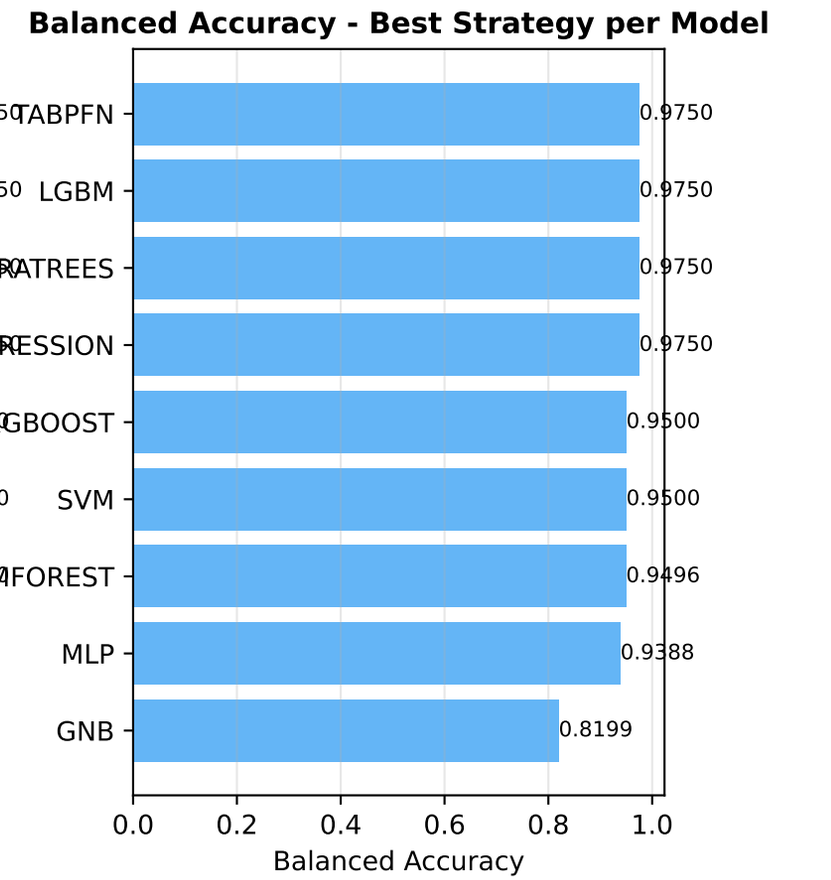
Model	Strategy	Accuracy	Balanced Acc	F1 (macro)	Log Loss
TABPFN	Original	0.975	0.975	0.9759725400457666	0.10301495360817092
TABPFN	SMOTE	0.975	0.975	0.9759725400457666	0.0914780965356532
TABPFN	RandomUnderSample	0.975	0.975	0.9759725400457666	0.09436706336408096
TABPFN	ClassWeights	0.975	0.975	0.9759725400457666	0.10301495360817092
LGBM	RandomOverSampler	0.975	0.975	0.9759725400457666	0.29103664419188646
EXTRATREES	Original	0.975	0.975	0.9759725400457666	0.1346915228864313
EXTRATREES	SMOTE	0.975	0.975	0.9759725400457666	0.13546829678899316
EXTRATREES	RandomOverSampler	0.975	0.975	0.9759725400457666	0.13332893419880792
LOGISTIC_REGRESSIOI	Original	0.975	0.975	0.9759725400457666	0.10206705744263189
LOGISTIC_REGRESSIOI	SMOTE	0.95	0.9522727272727271	0.9522727272727271	0.0940890950444778 2

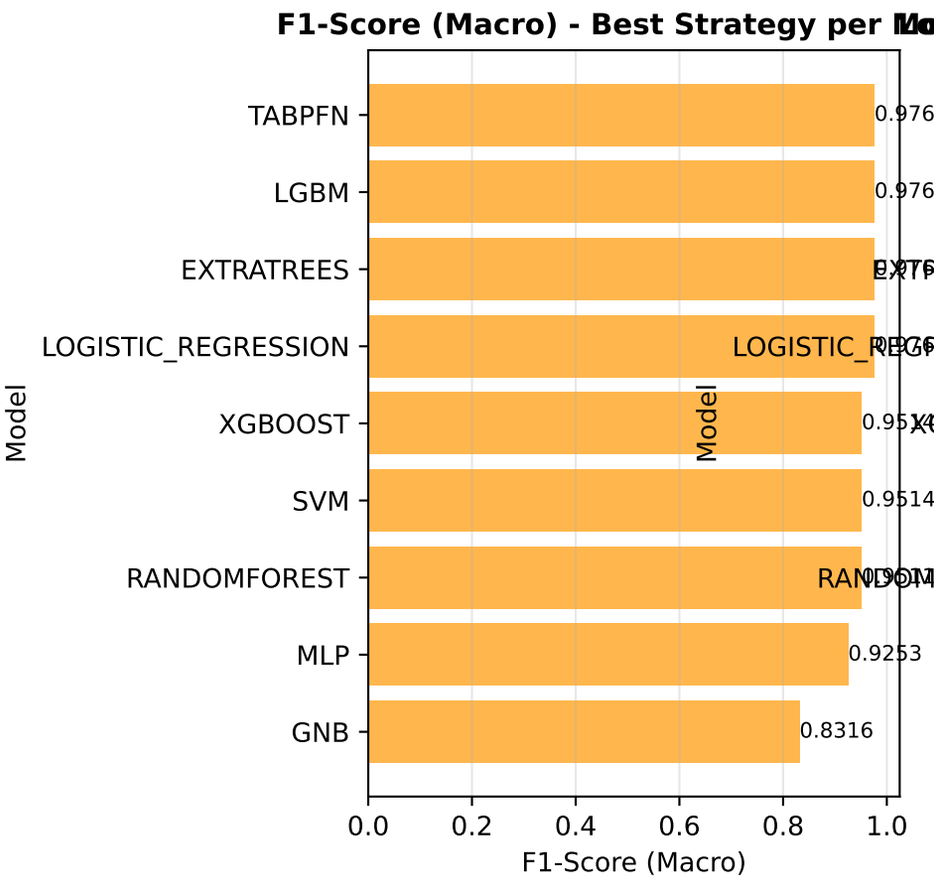
Best Strategy per Model (by Balanced Accuracy)

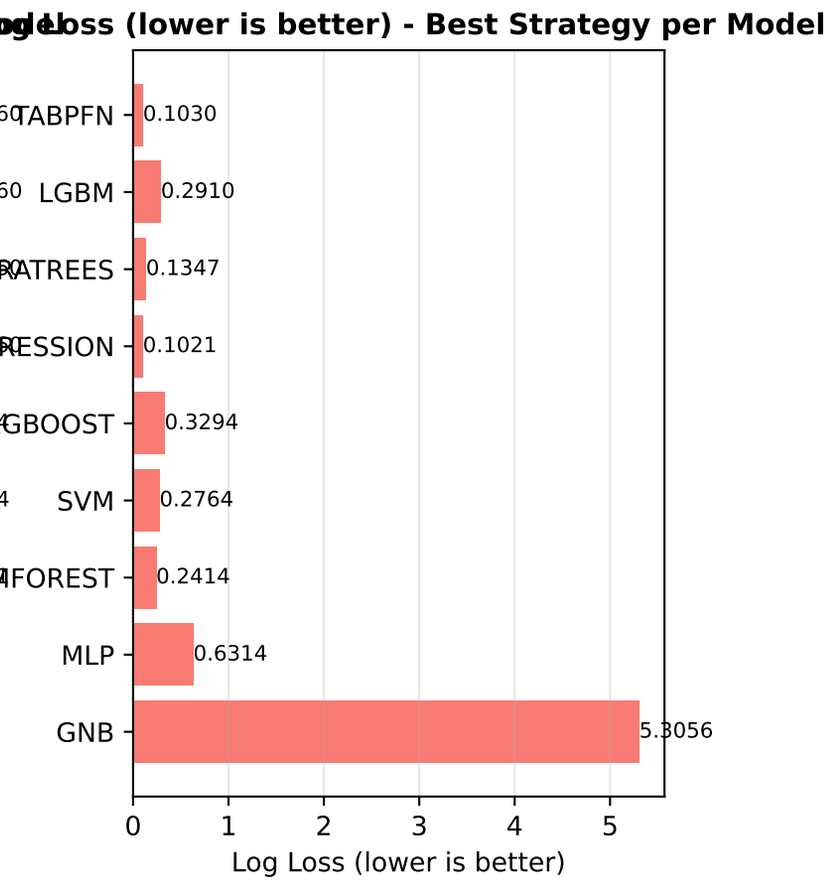
Model	Strategy	Accuracy	Balanced Acc	F1 (macro)	Log Loss
EXTRATREES	Original	0.975	0.975	0.9759725400457666	0.1346915228864313
LGBM	RandomOverSampler	0.975	0.975	0.9759725400457666	0.29103664419188646
LOGISTIC_REGRESSIOI	Original	0.975	0.975	0.9759725400457666	0.10206705744263189
TABPFN	Original	0.975	0.975	0.9759725400457666	0.10301495360817092
SVM	SMOTE	0.95	0.95	0.9513888888888888	0.27642791493445307
XGBOOST	Original	0.95	0.95	0.9513888888888888	0.3293717856724081
RANDOMFOREST	SVM SMOTE	0.95	0.9495798319327731	0.9511060834590246	0.24142813041517344
MLP	RandomOverSampler	0.925	0.9388111888111889	0.9253038628038628	0.6313526015234503
GNB	Original	0.825	0.8198717948717948	0.8315960355434045	5.305609367390395

Performance Metrics Comparison

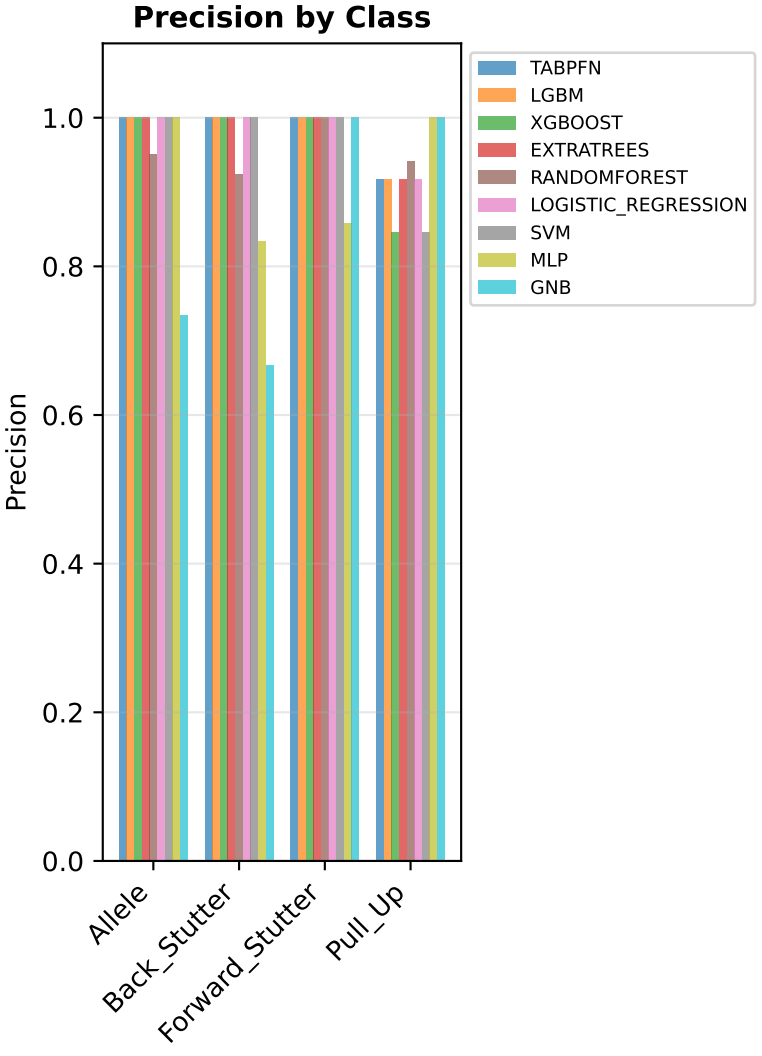


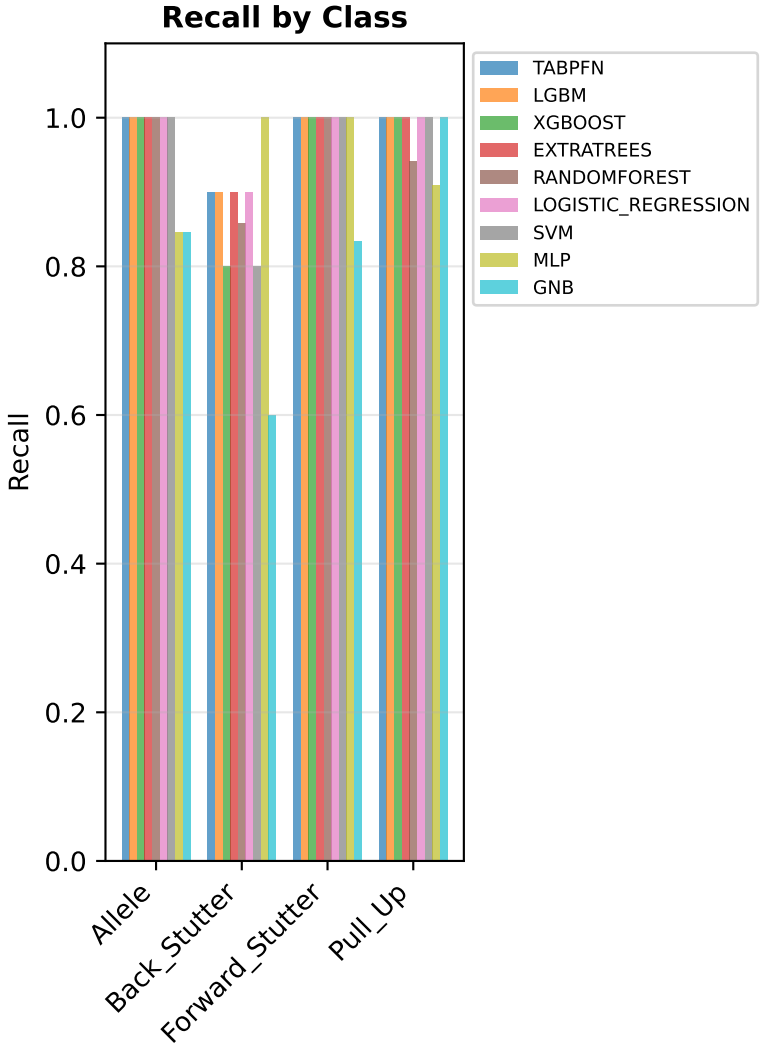


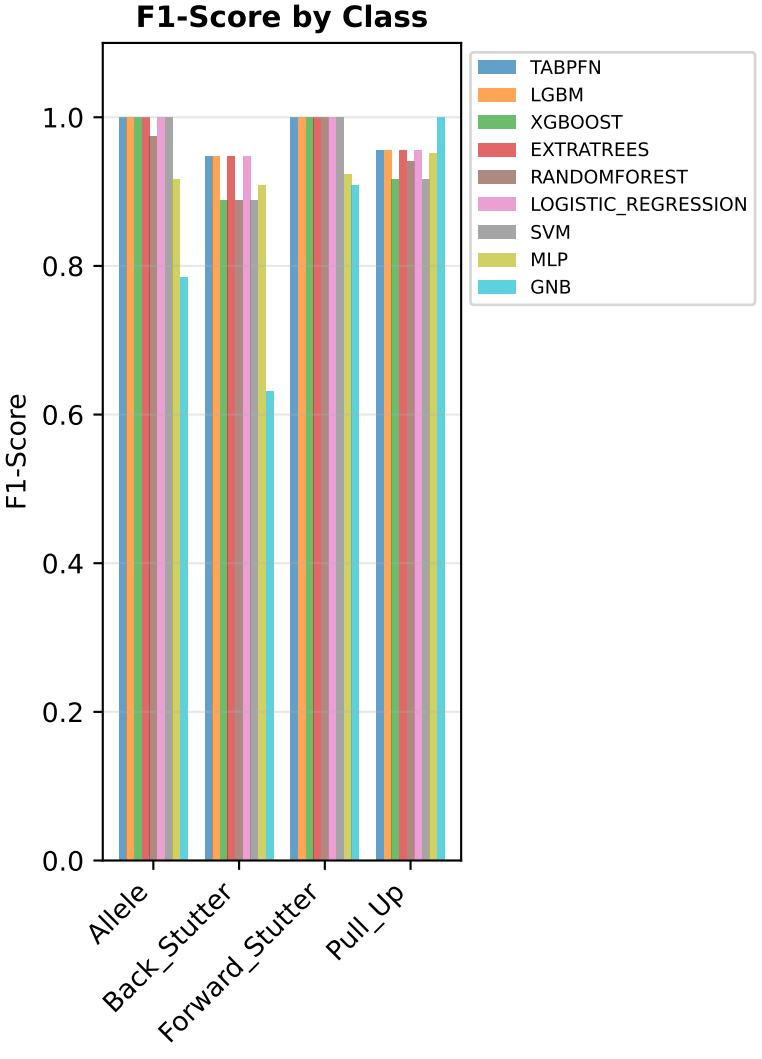


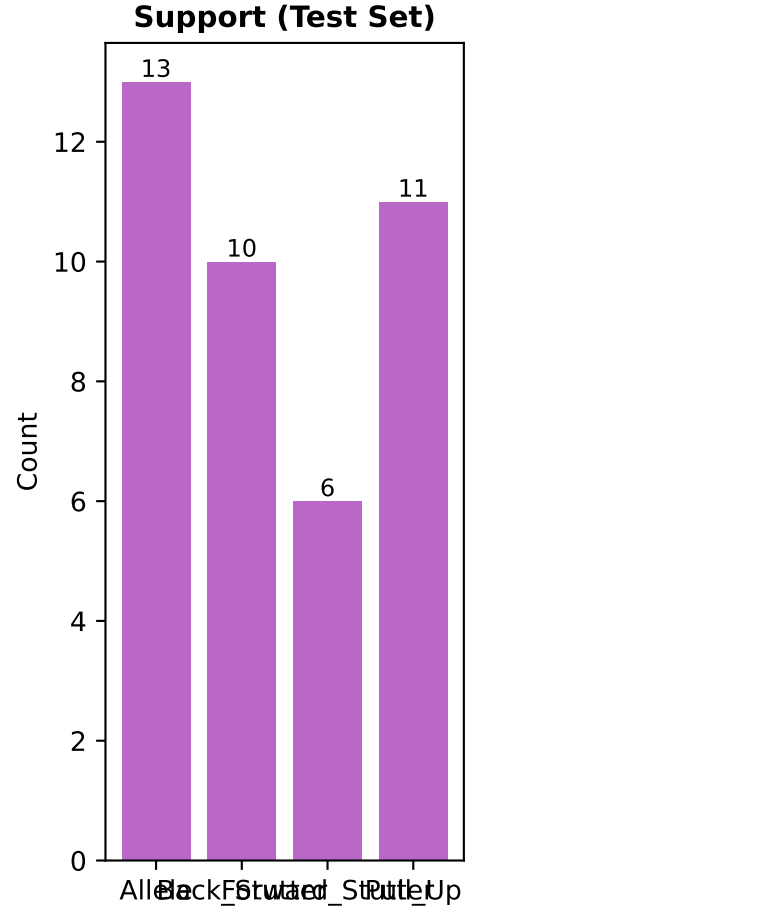


Per-Class Performance Comparison (Best Strategy per Model)







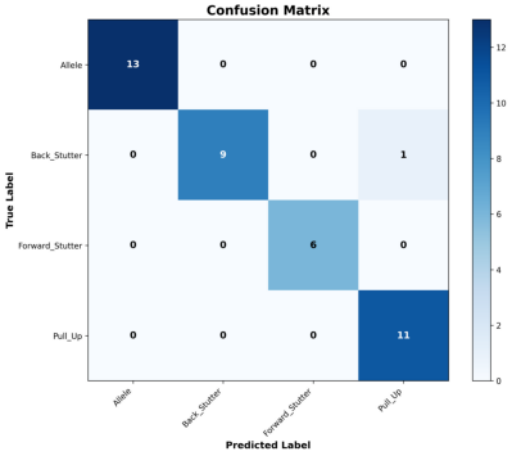


TABPFN - Original Strategy

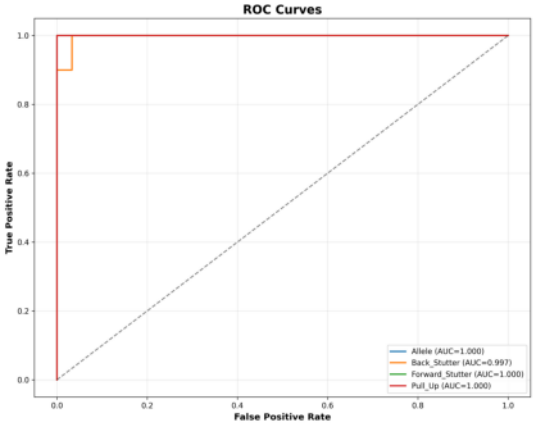
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1030

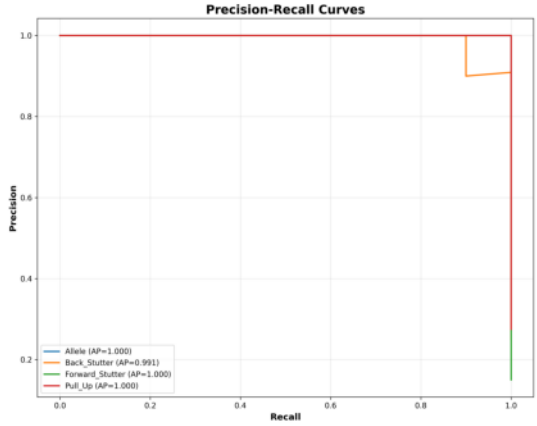
Confusion Matrix



ROC Curves



Precision-Recall Curves

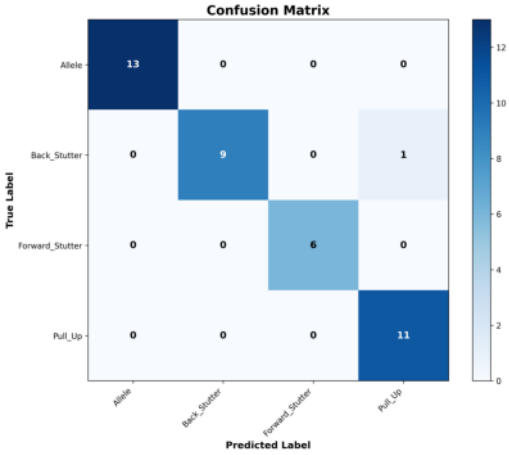


TABPFN - SMOTE Strategy

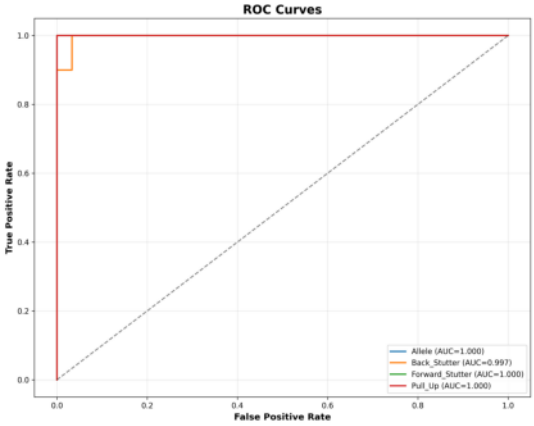
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.0915

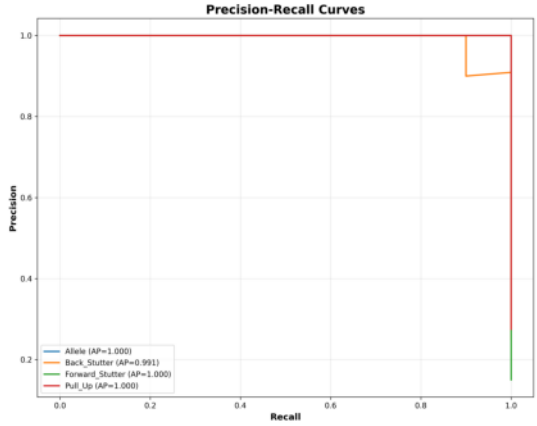
Confusion Matrix



ROC Curves



Precision-Recall Curves

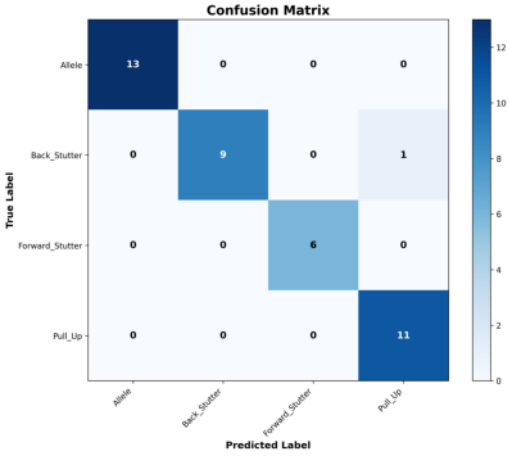


TABPFN - RandomUnderSampler Strategy

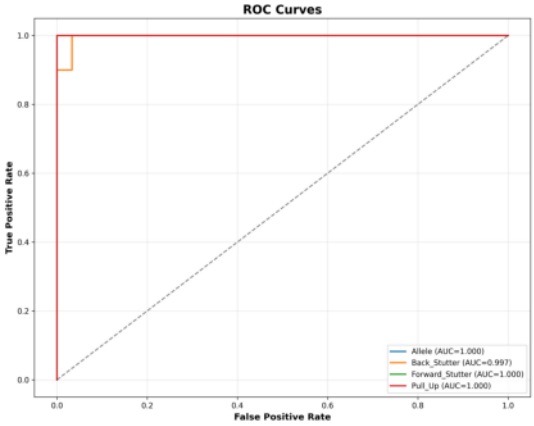
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.0944

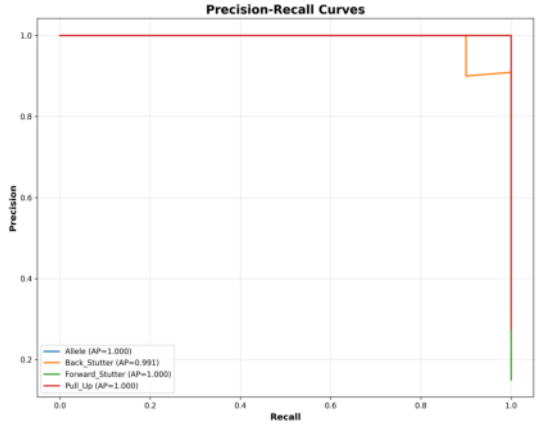
Confusion Matrix



ROC Curves



Precision-Recall Curves

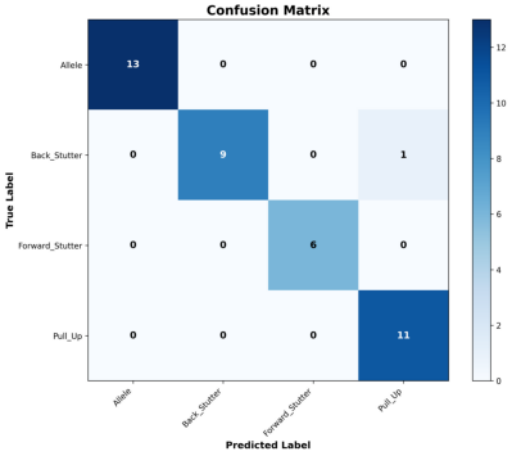


TABPFN - ClassWeights Strategy

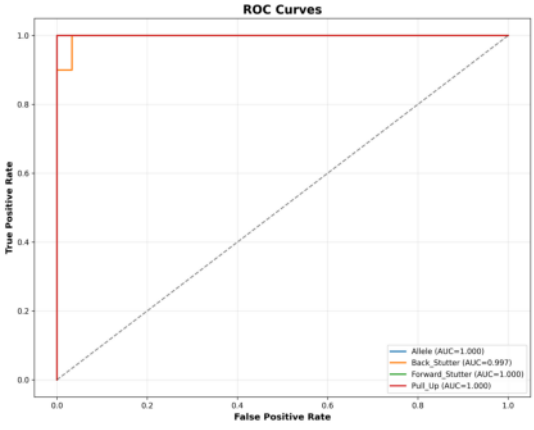
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1030

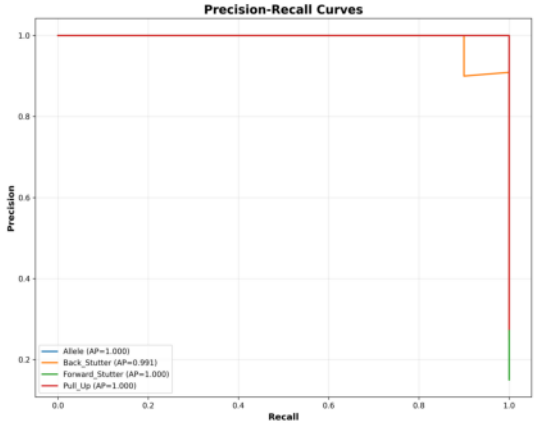
Confusion Matrix



ROC Curves



Precision-Recall Curves

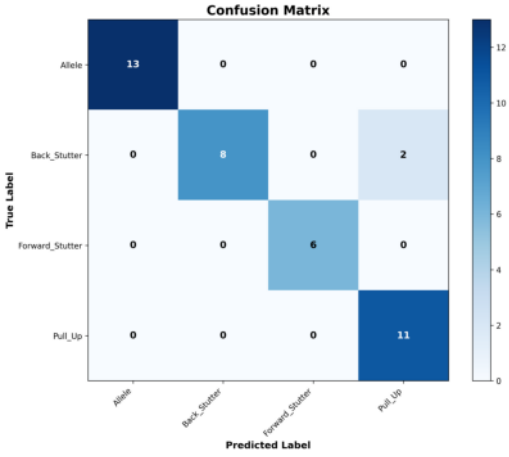


LGBM - Original Strategy

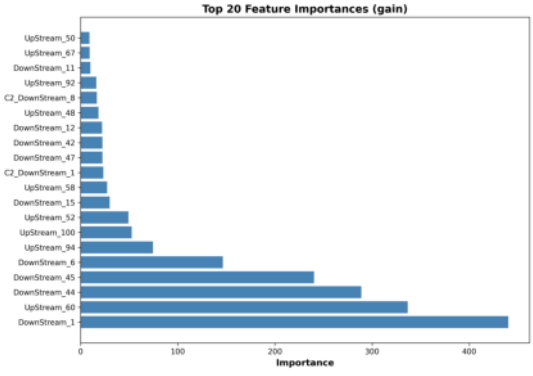
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.3934

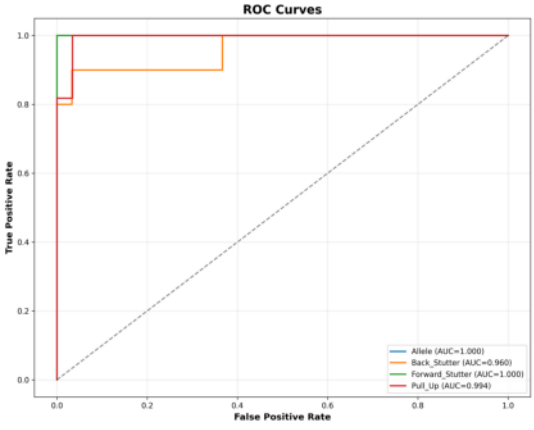
Confusion Matrix



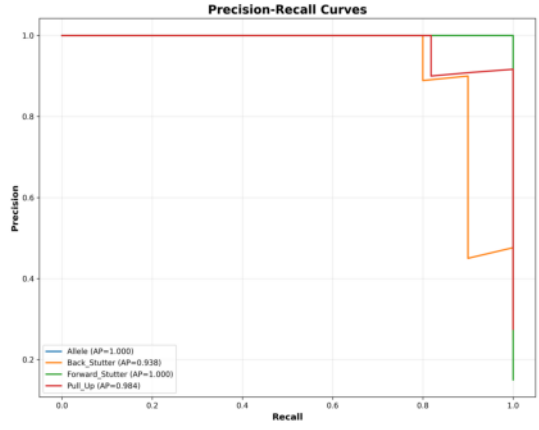
Feature Importance



ROC Curves

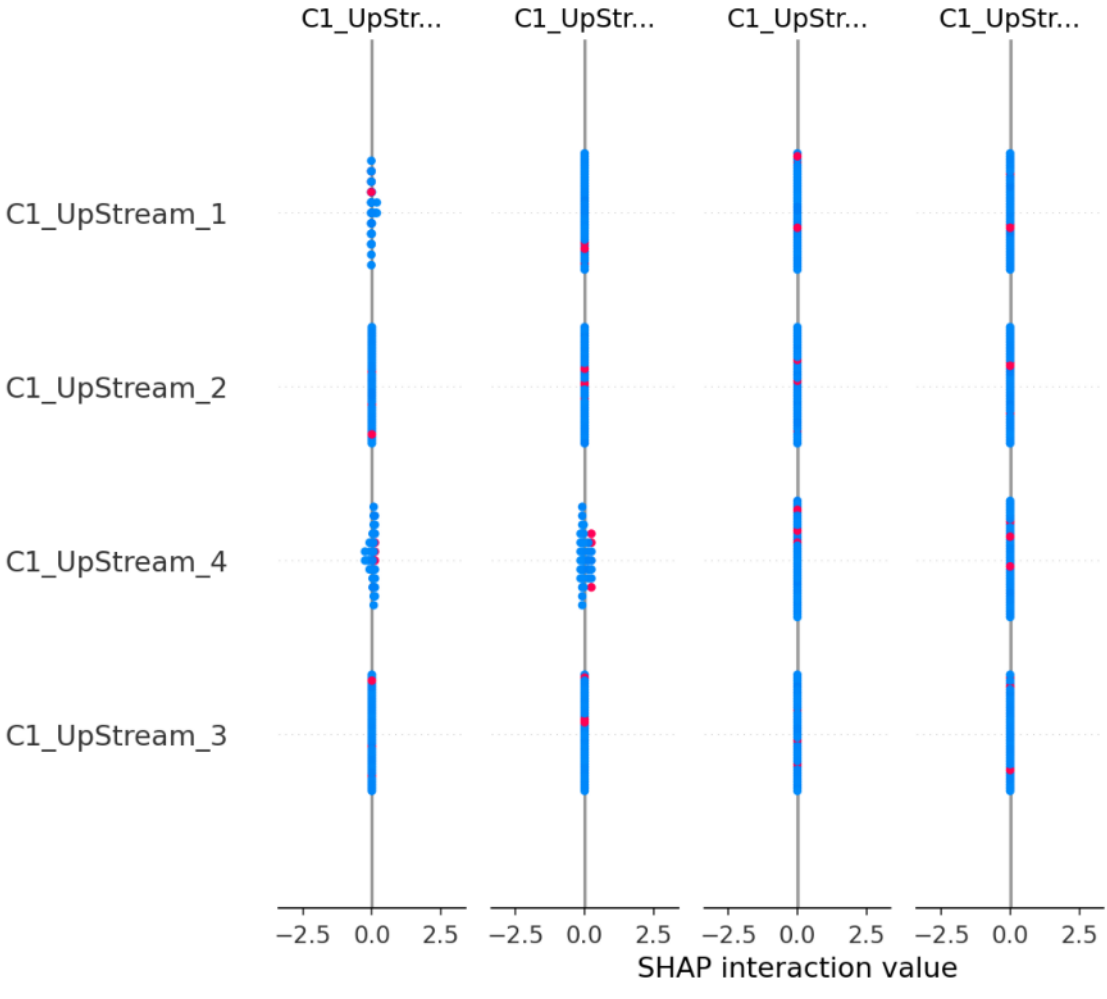


Precision-Recall Curves

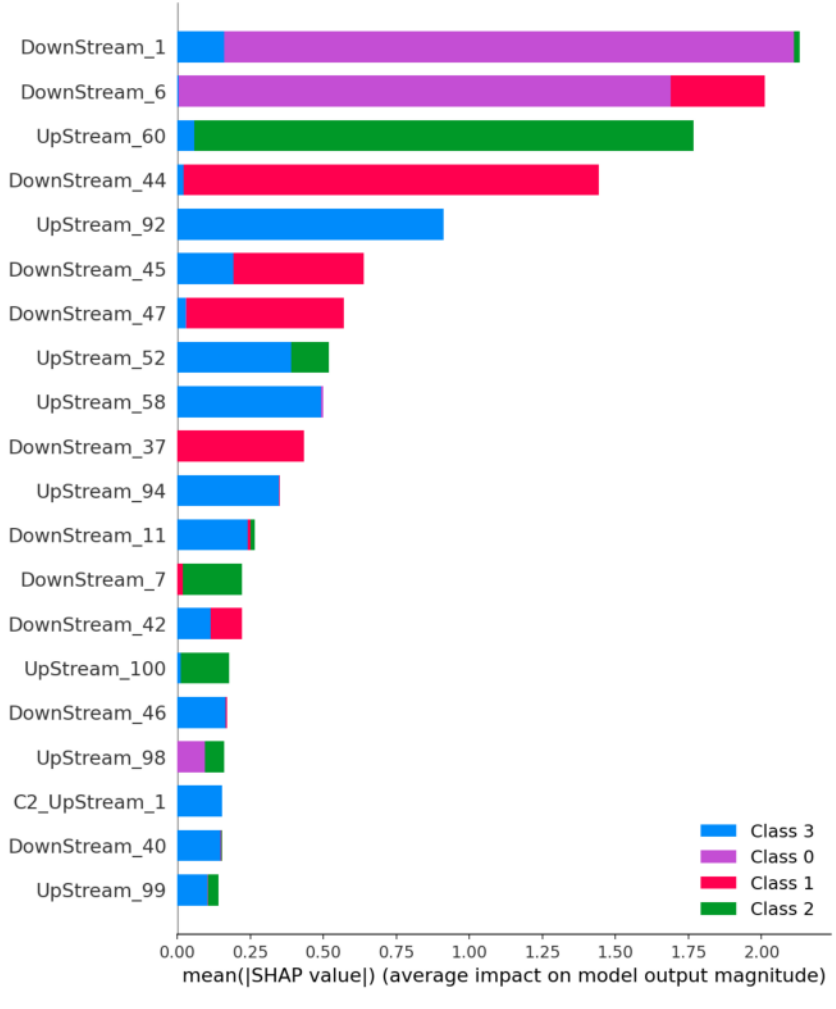


LGBM - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

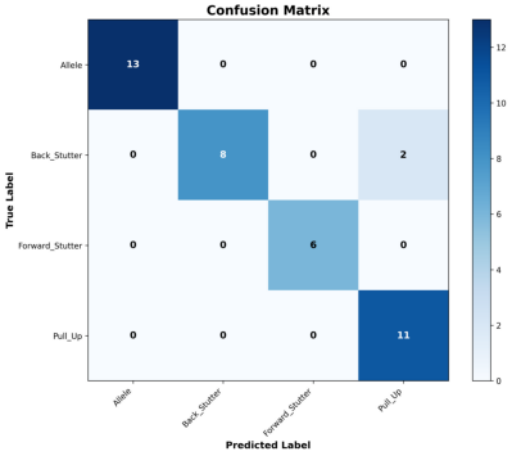


LGBM - SMOTE Strategy

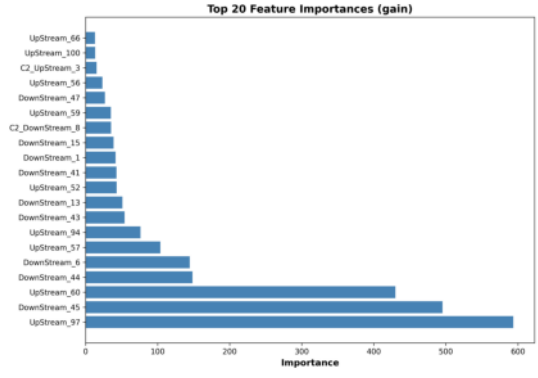
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.3684

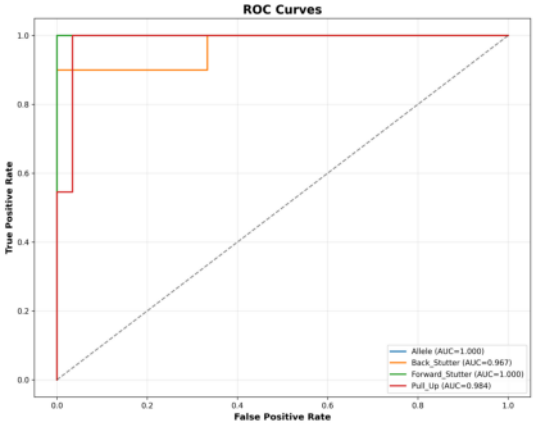
Confusion Matrix



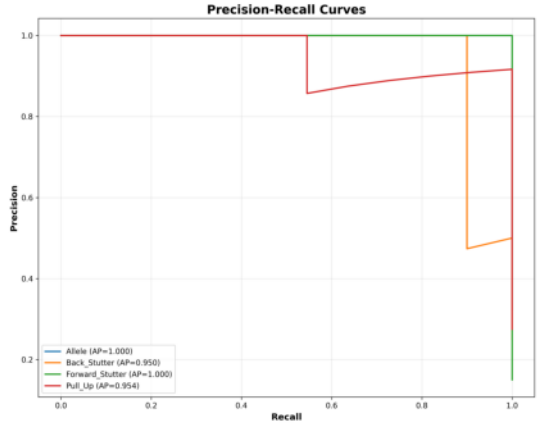
Feature Importance



ROC Curves

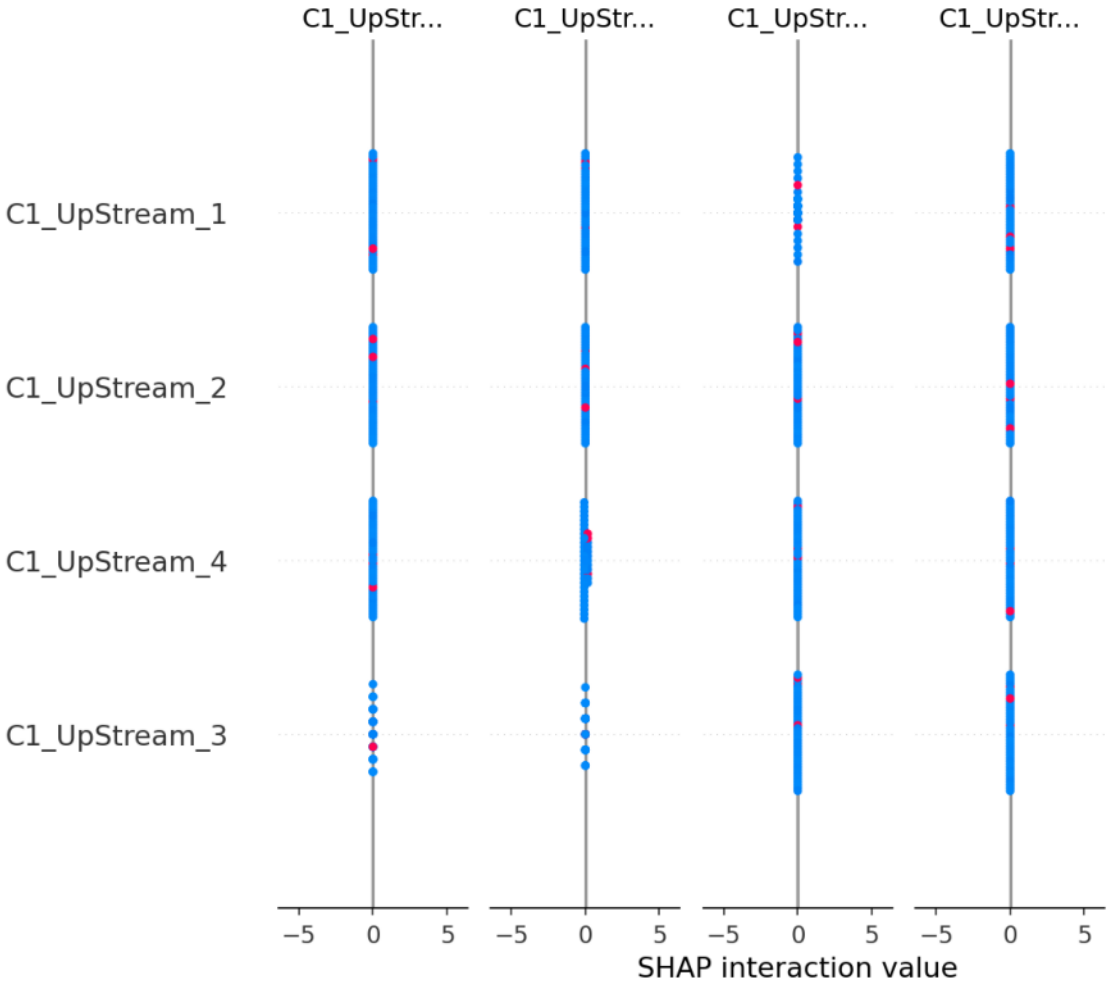


Precision-Recall Curves

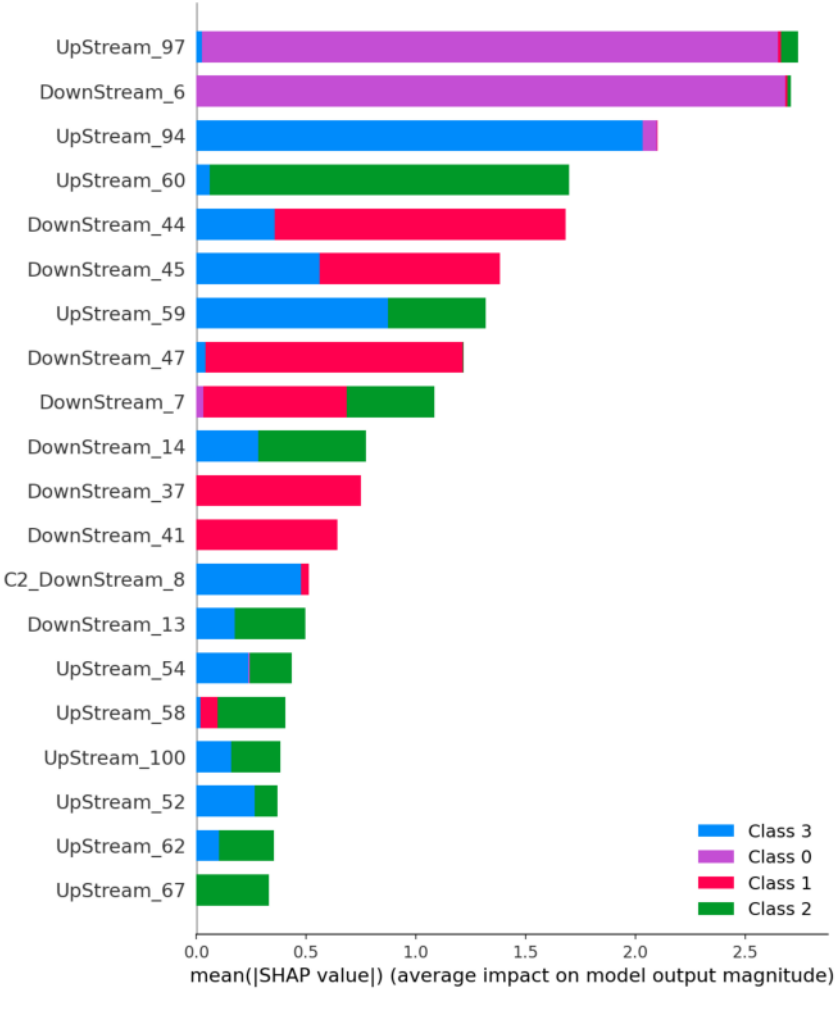


LGBM - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

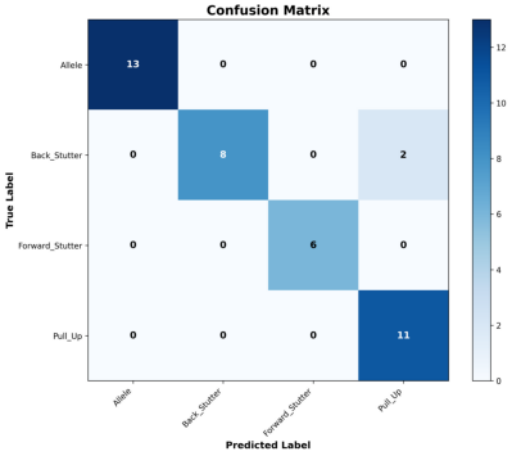


LGBM - RandomOverSampler Strategy

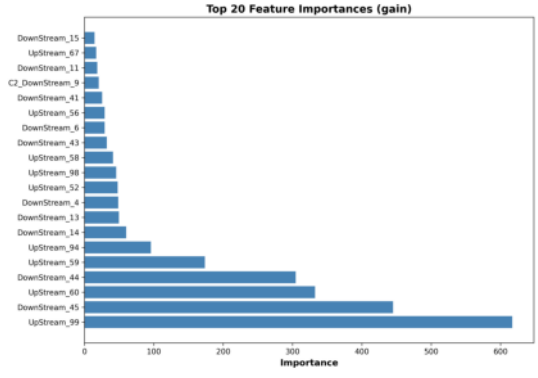
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.2910

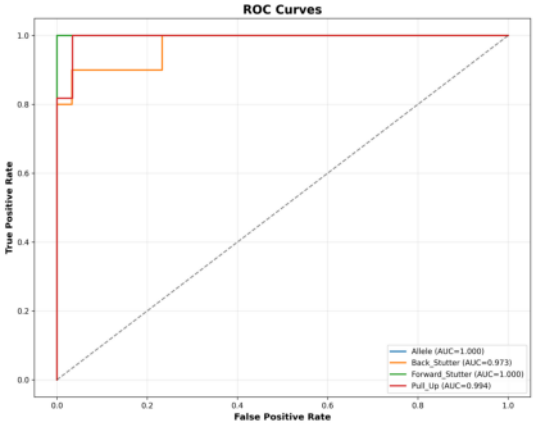
Confusion Matrix



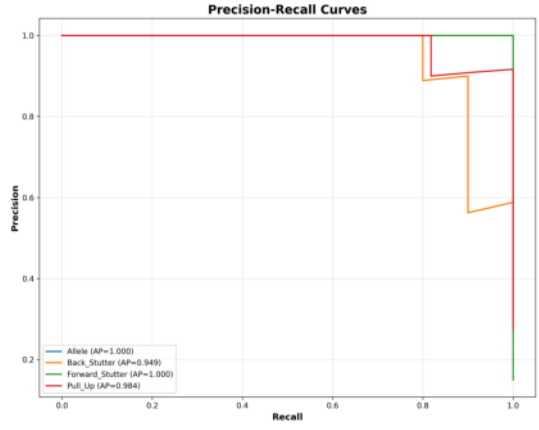
Feature Importance



ROC Curves

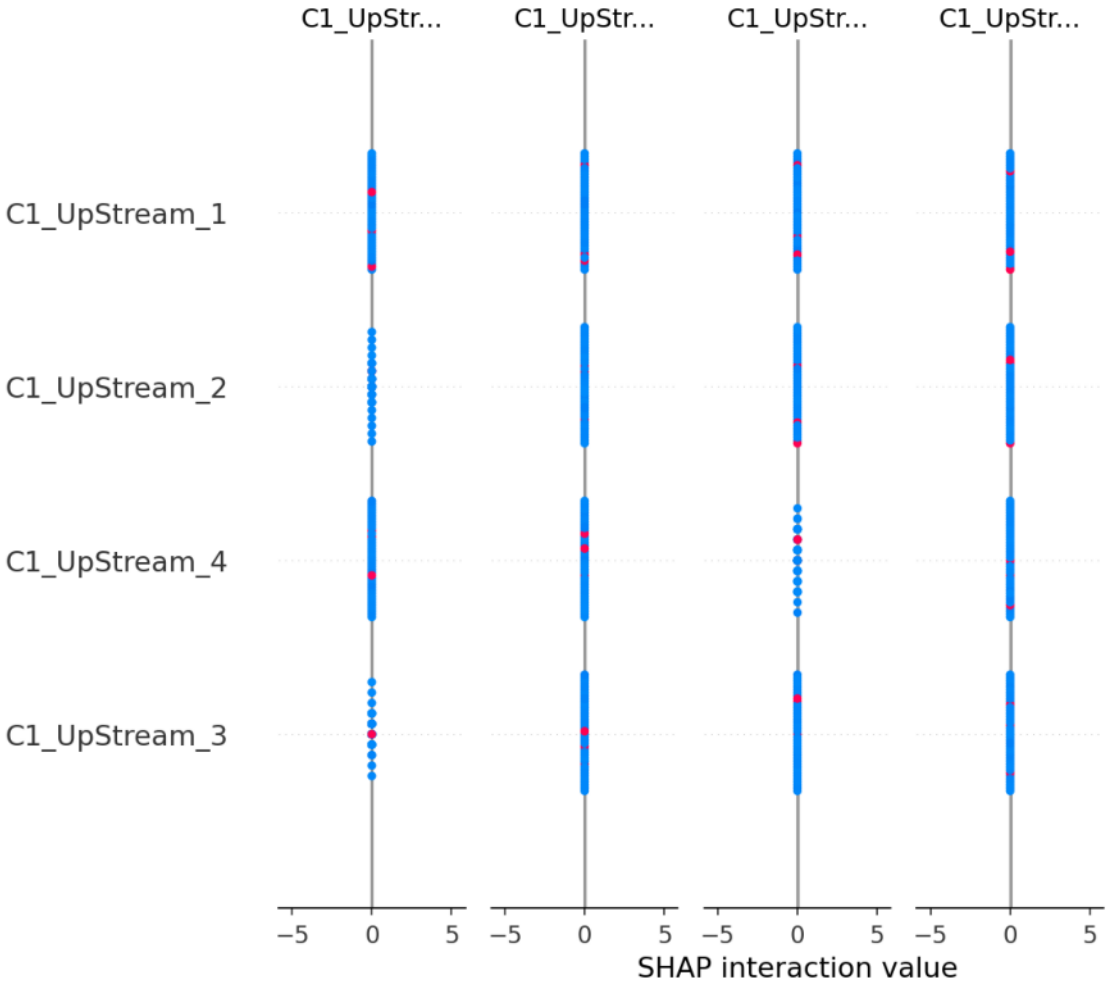


Precision-Recall Curves

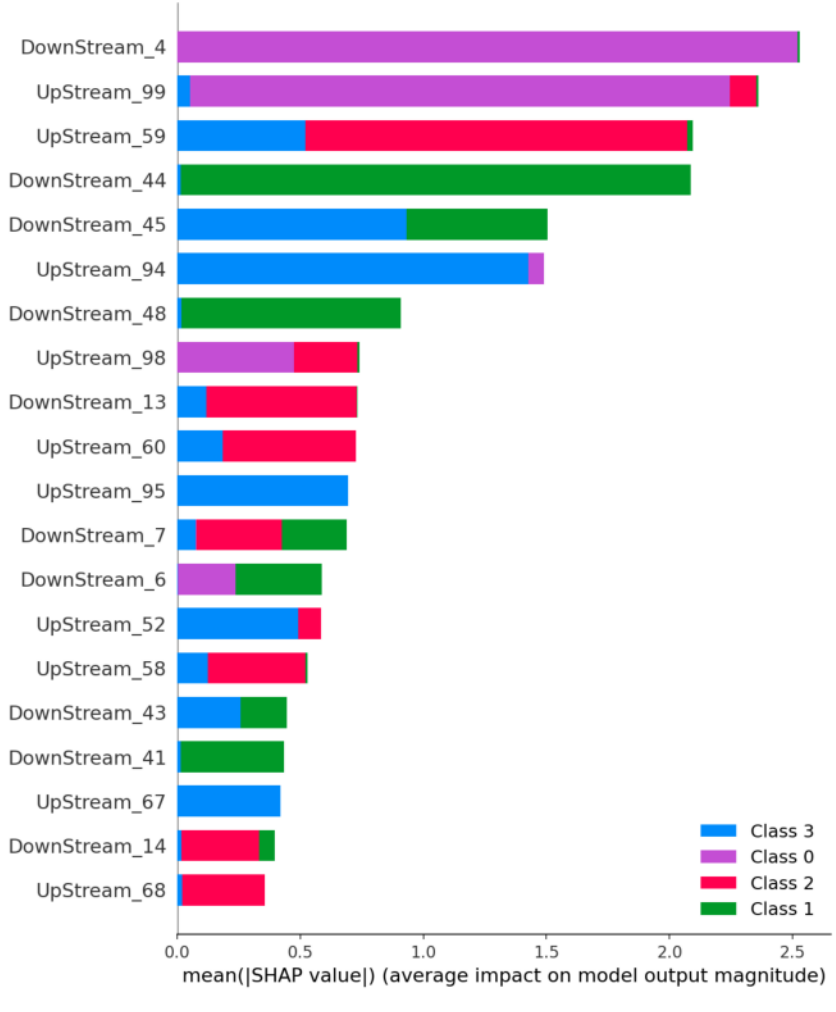


LGBM - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

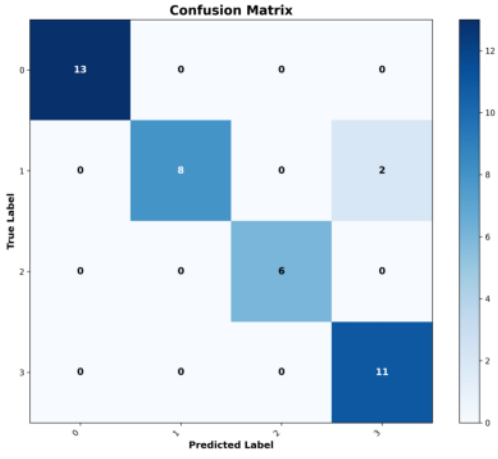


XGBOOST - Original Strategy

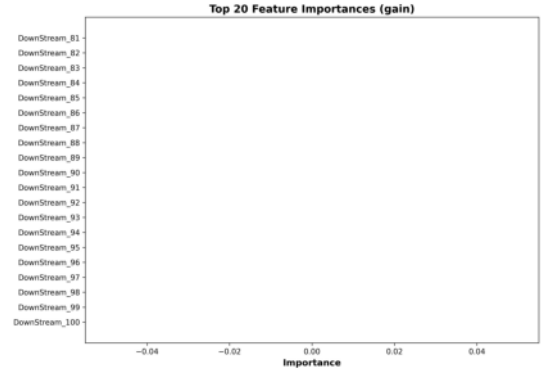
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.3294

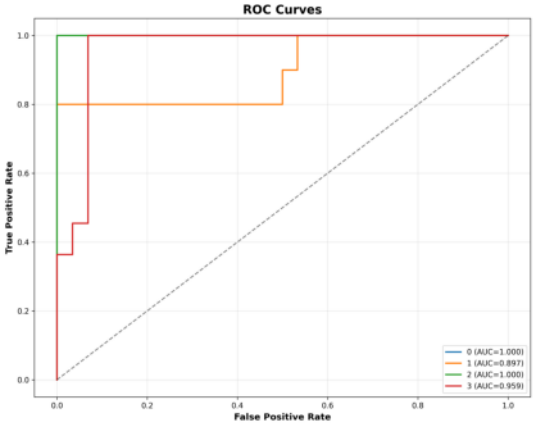
Confusion Matrix



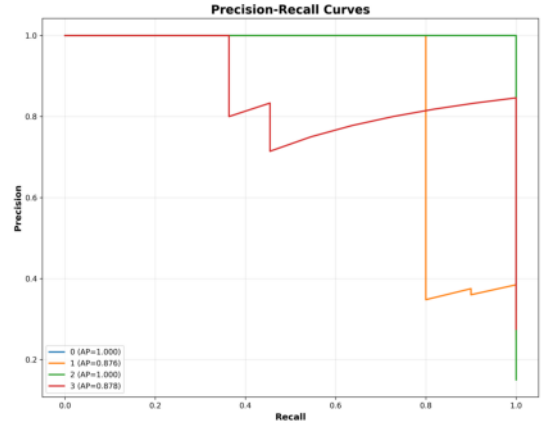
Feature Importance



ROC Curves

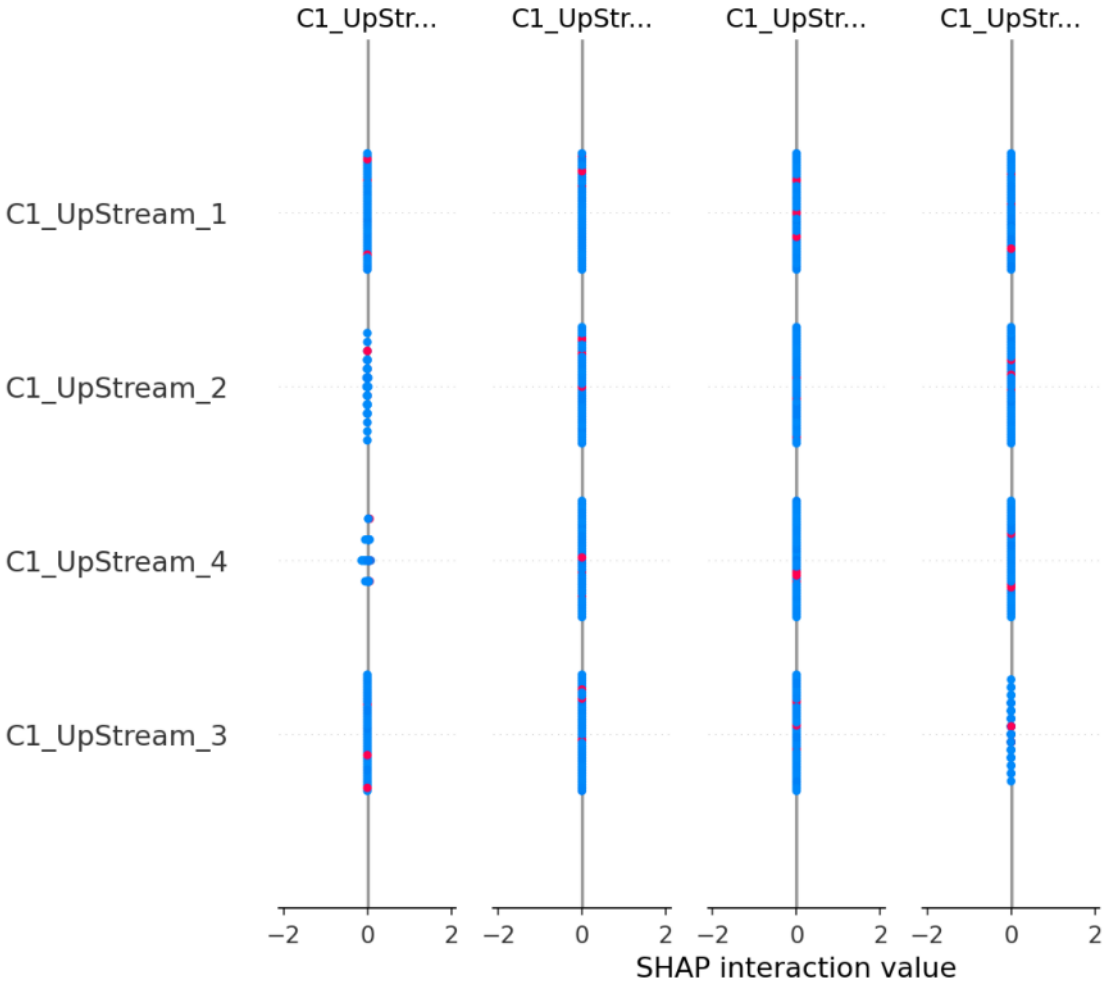


Precision-Recall Curves

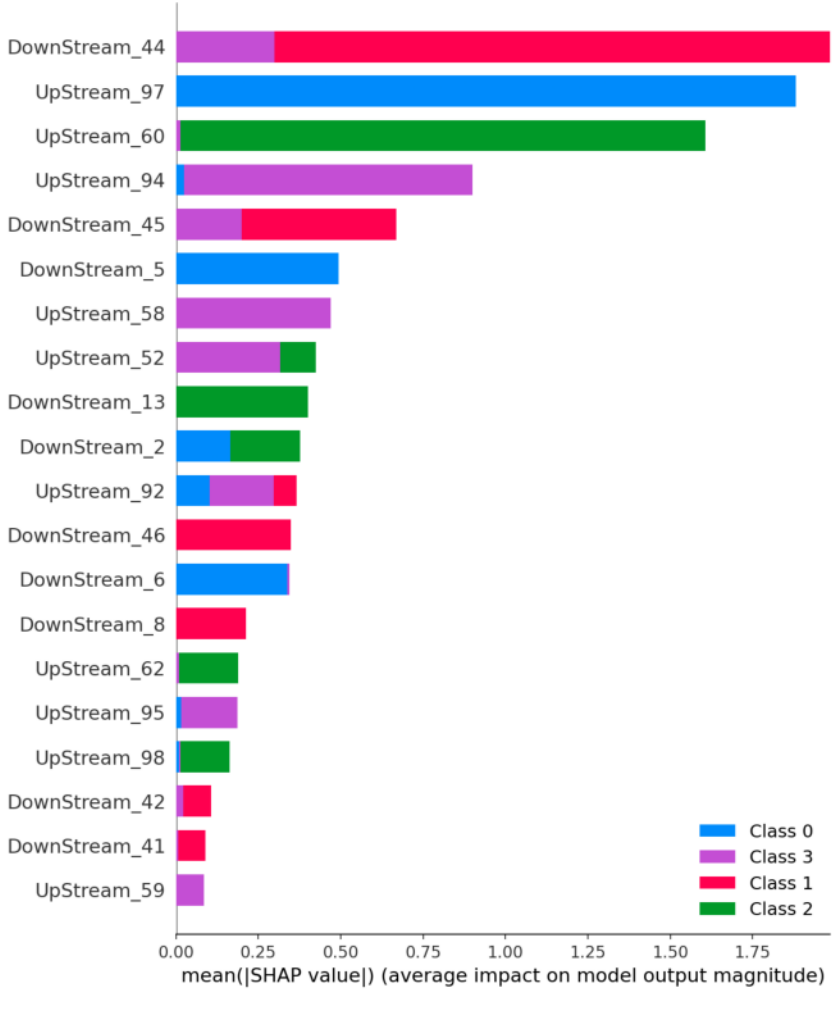


XGBOOST - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

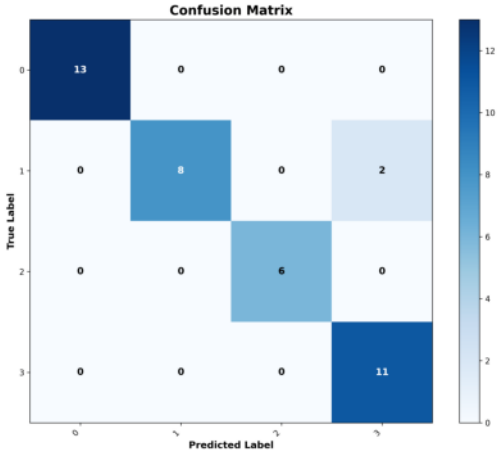


XGBOOST - SMOTE Strategy

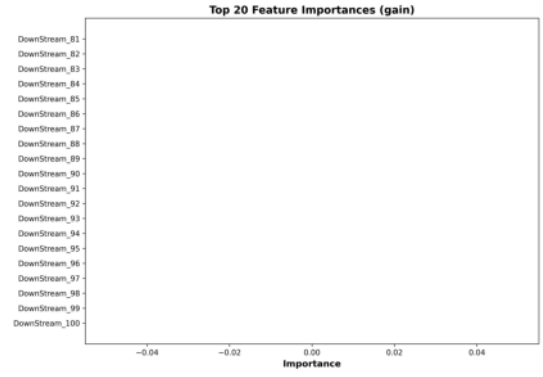
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.3301

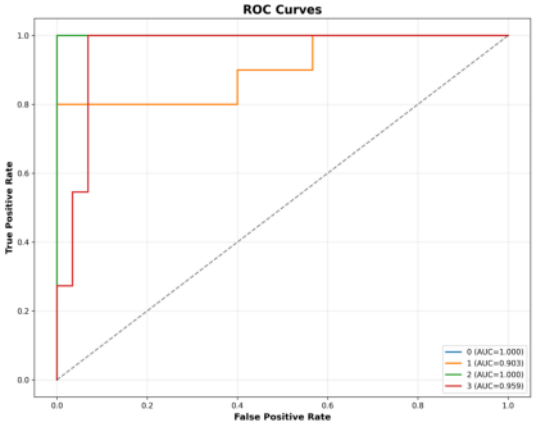
Confusion Matrix



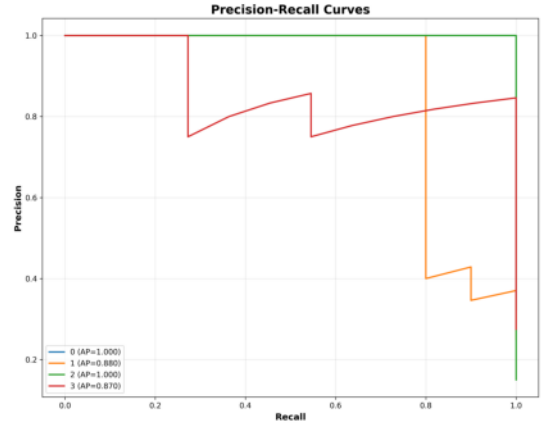
Feature Importance



ROC Curves

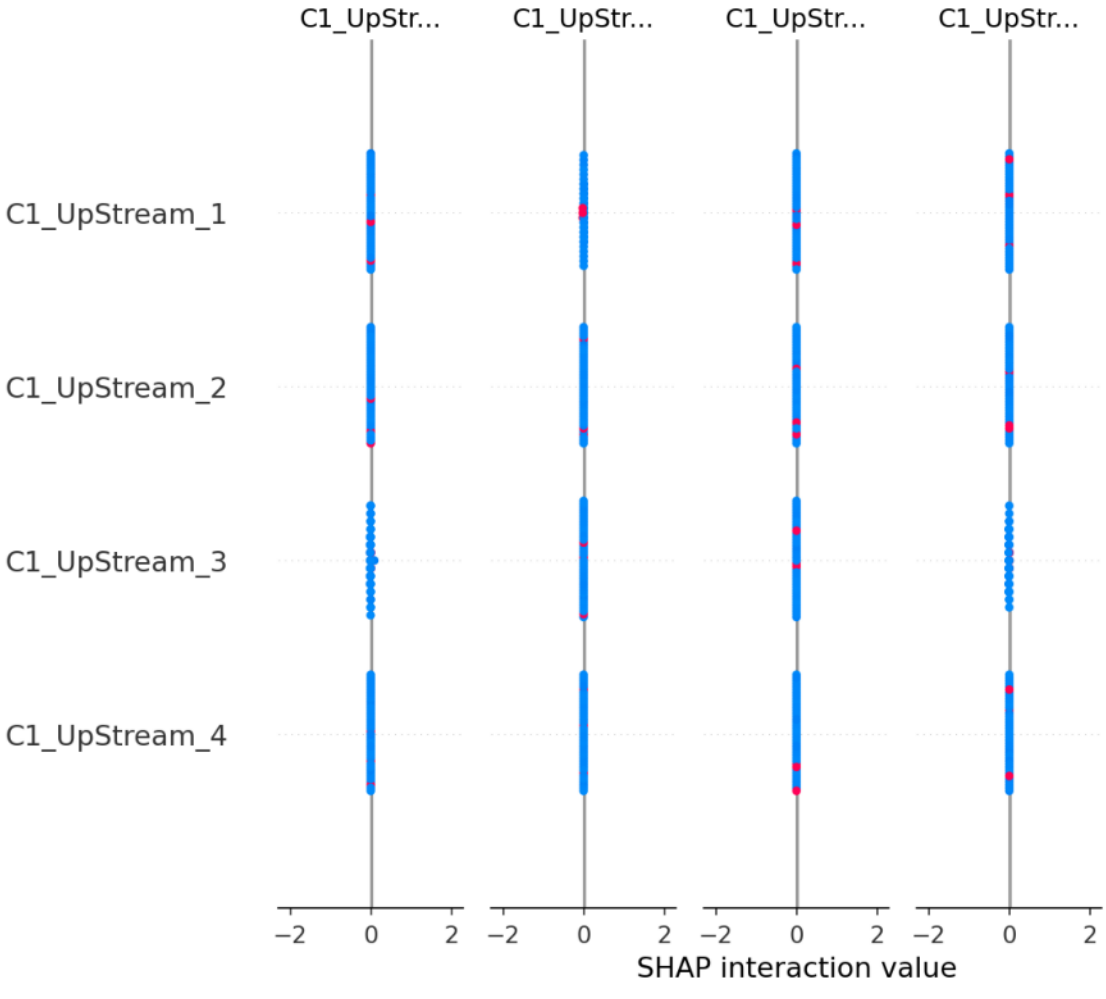


Precision-Recall Curves

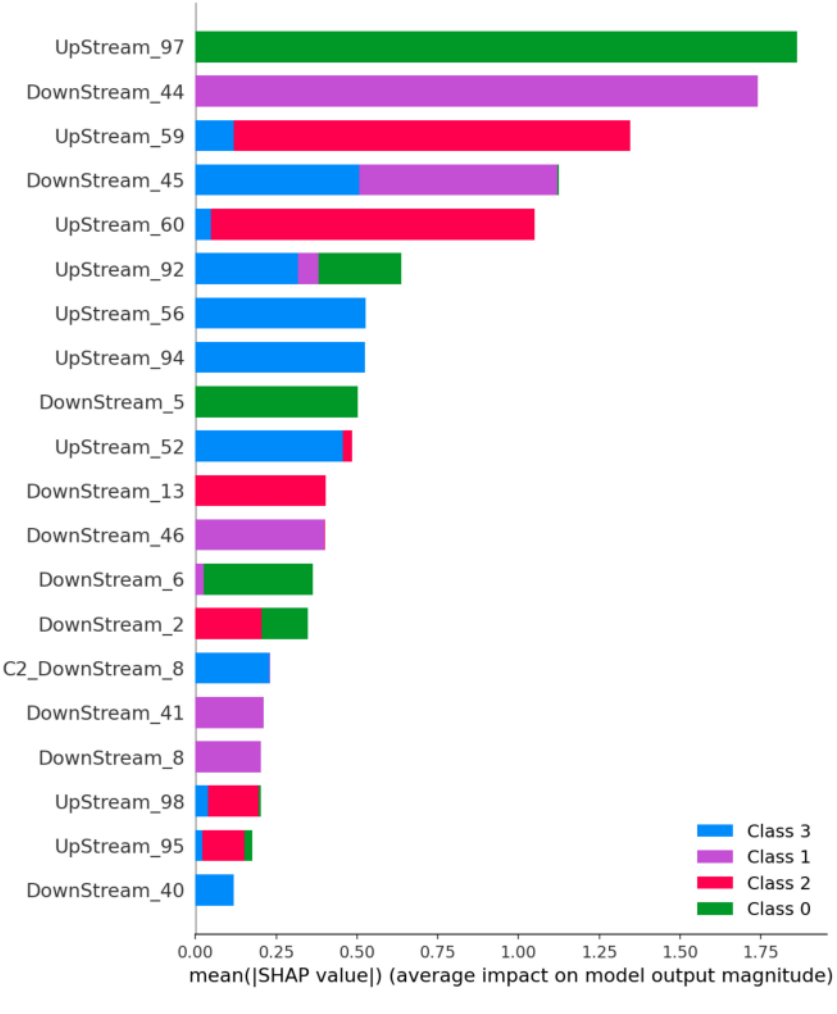


XGBOOST - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

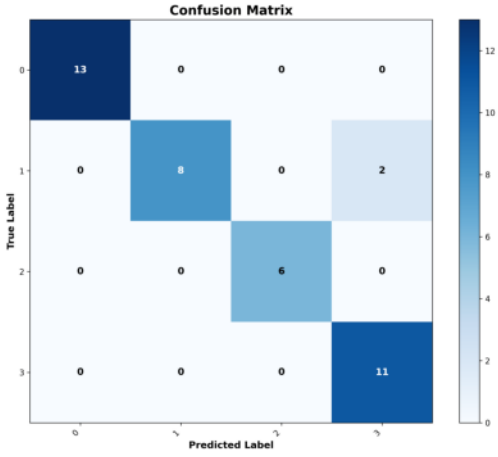


XGBOOST - RandomOverSampler Strategy

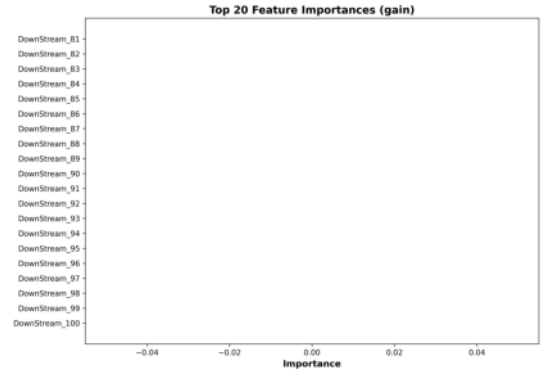
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.3132

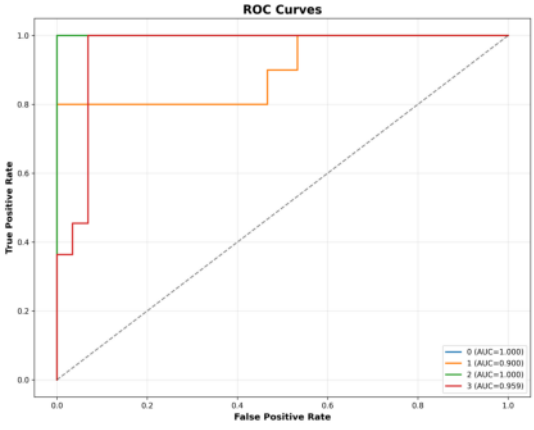
Confusion Matrix



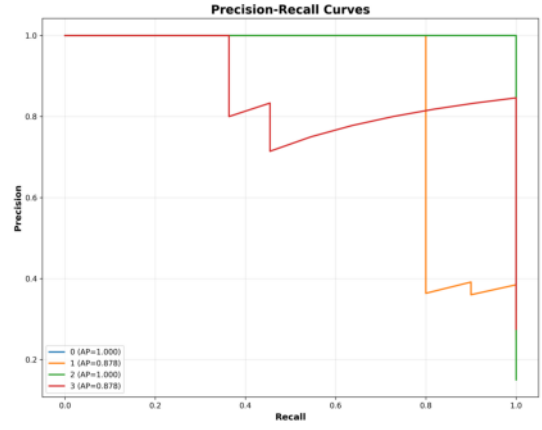
Feature Importance



ROC Curves

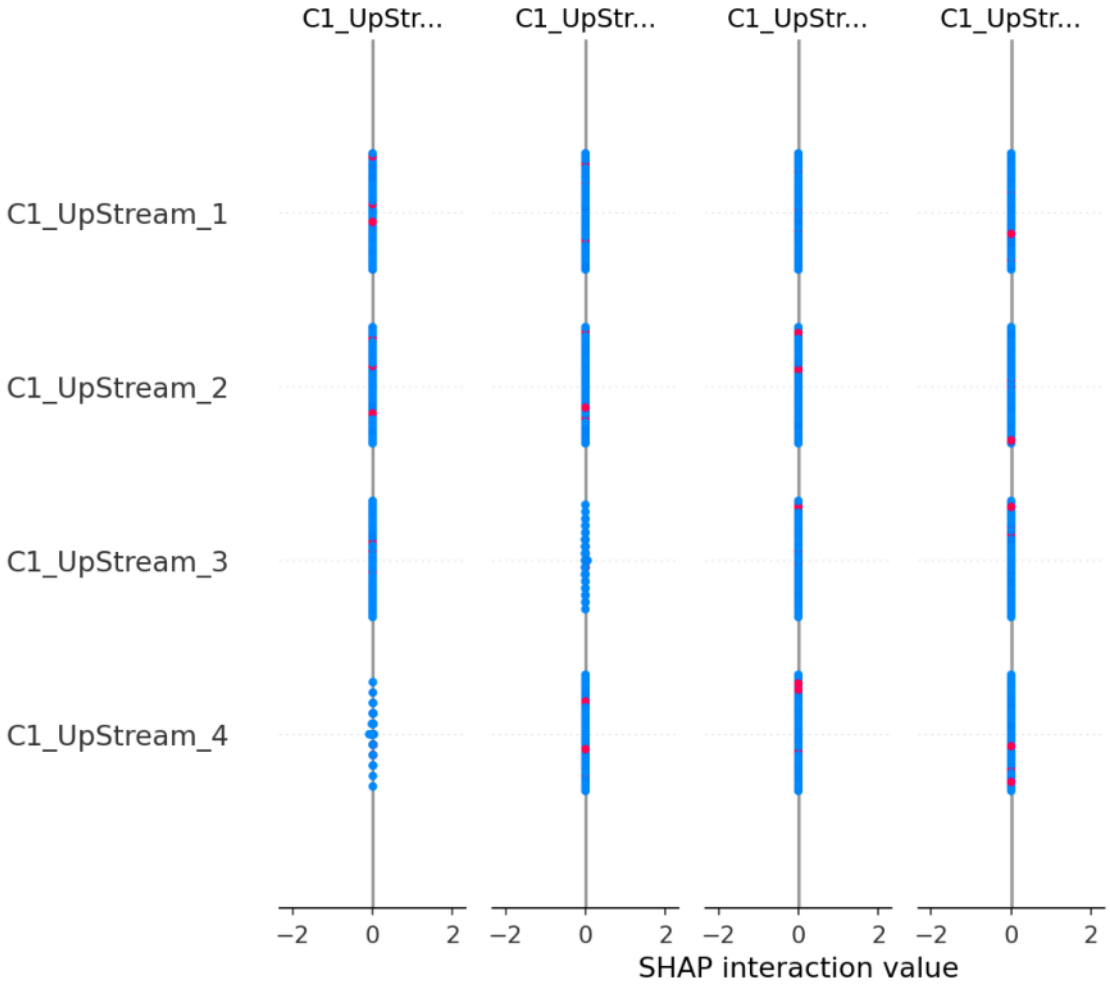


Precision-Recall Curves

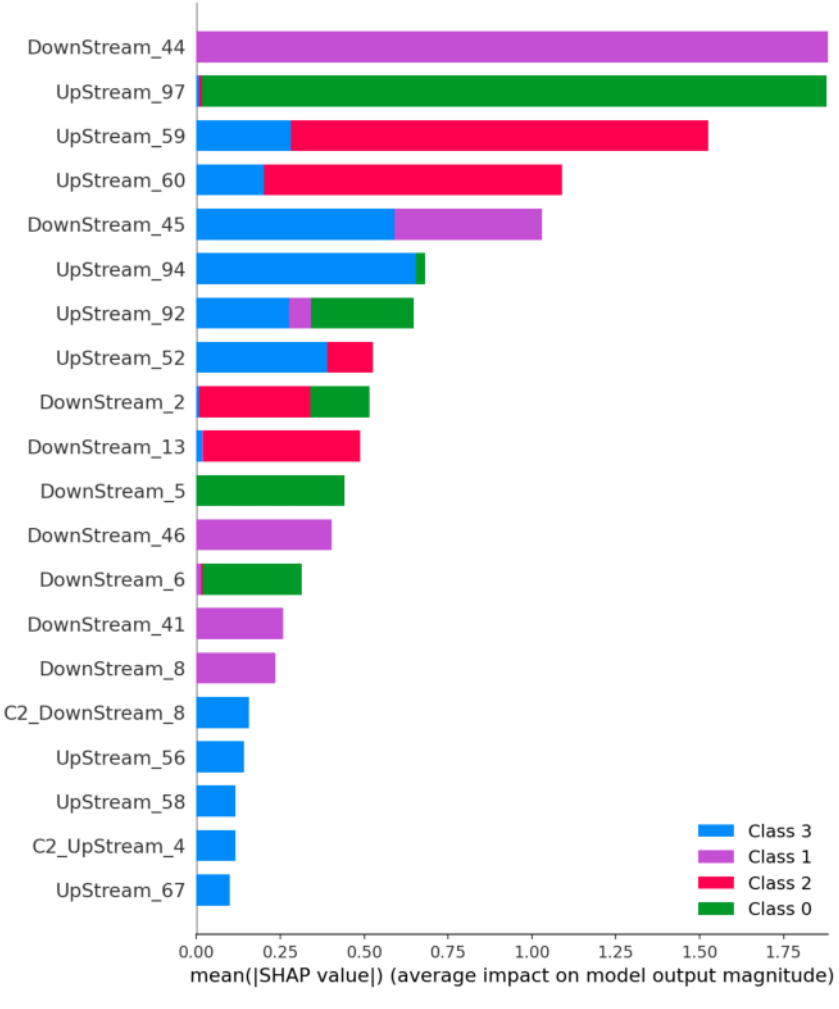


XGBOOST - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

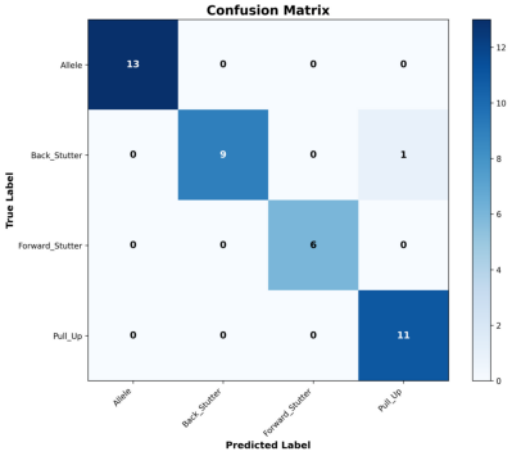


EXTRATREES - Original Strategy

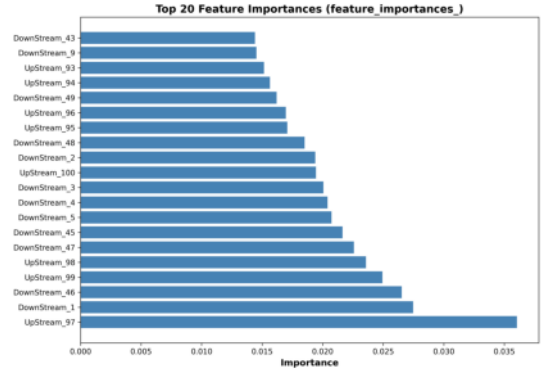
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1347

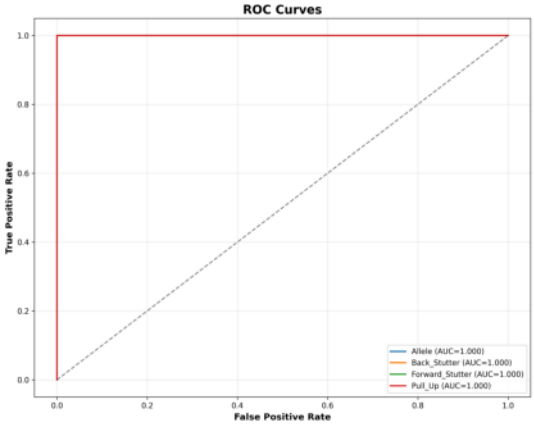
Confusion Matrix



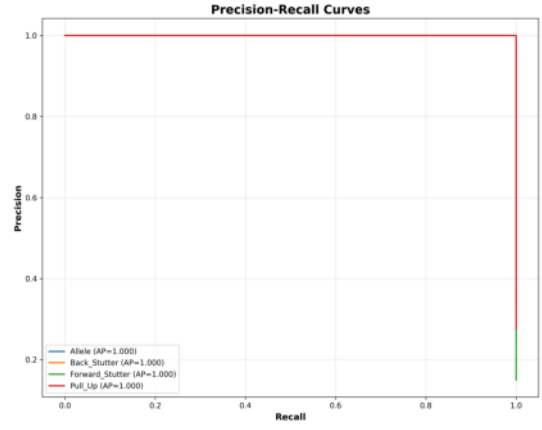
Feature Importance



ROC Curves



Precision-Recall Curves

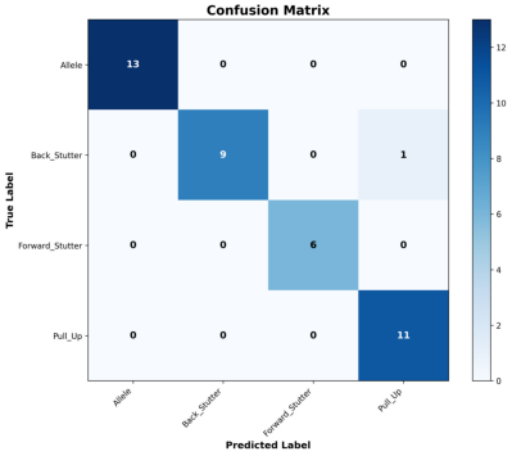


EXTRATREES - SMOTE Strategy

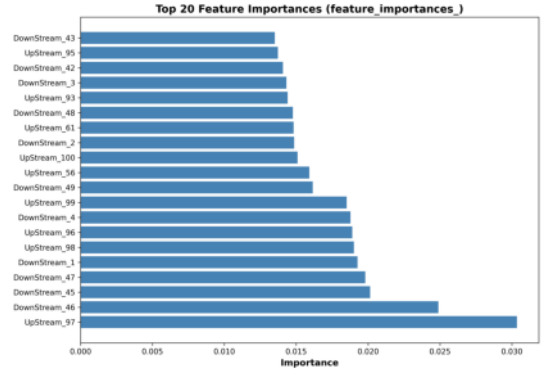
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1355

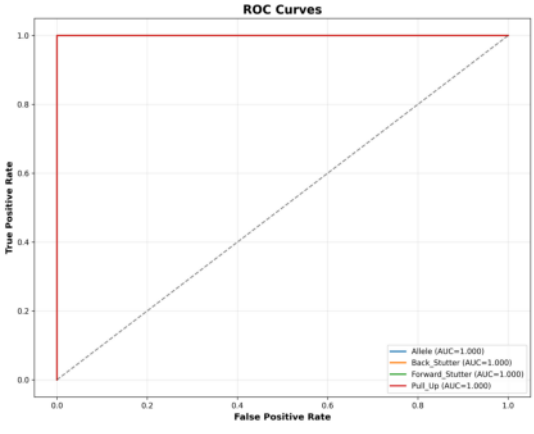
Confusion Matrix



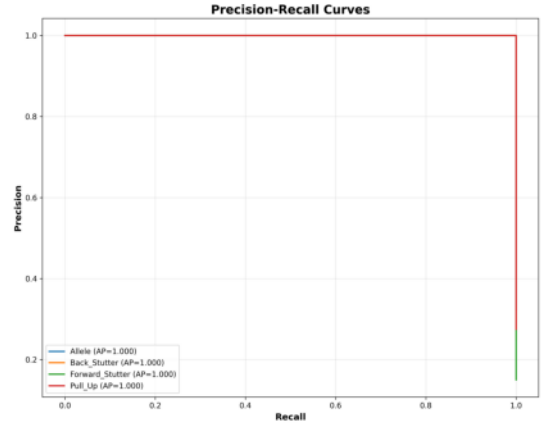
Feature Importance



ROC Curves



Precision-Recall Curves

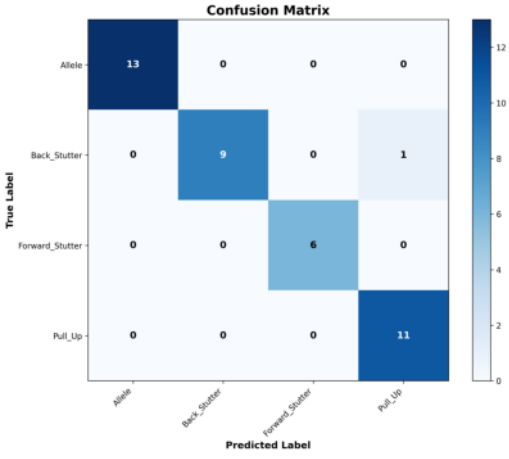


EXTRATREES - RandomOverSampler Strategy

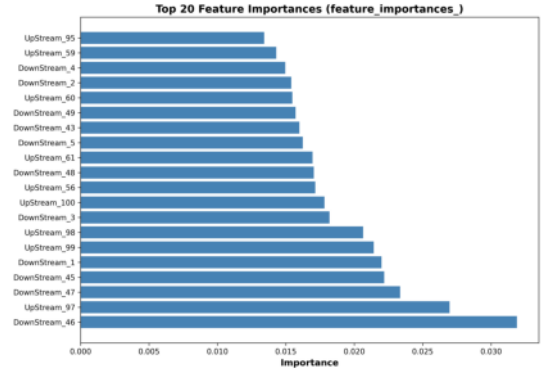
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1333

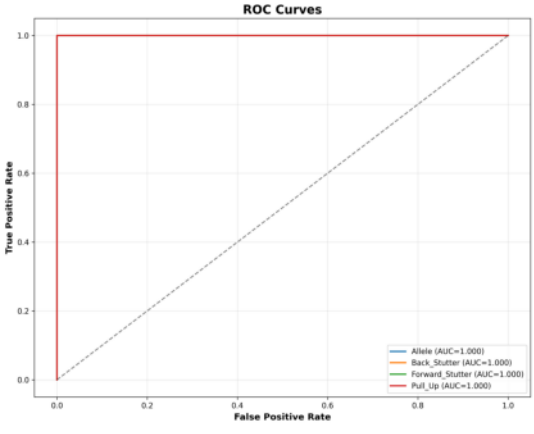
Confusion Matrix



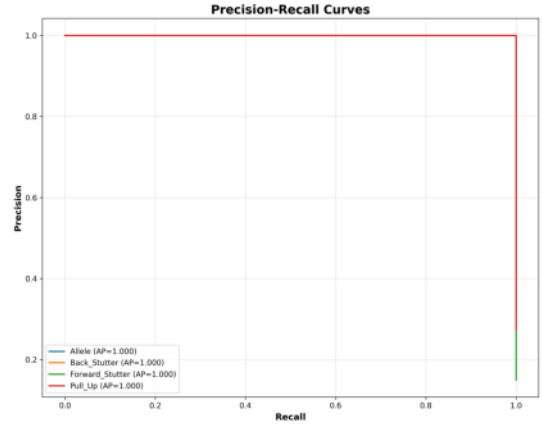
Feature Importance



ROC Curves



Precision-Recall Curves

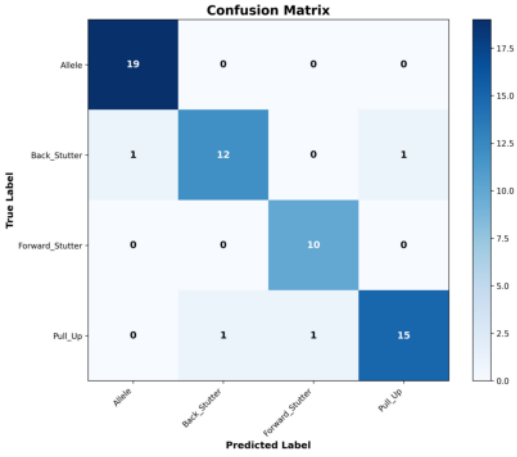


RANDOMFOREST - No Resampling Strategy

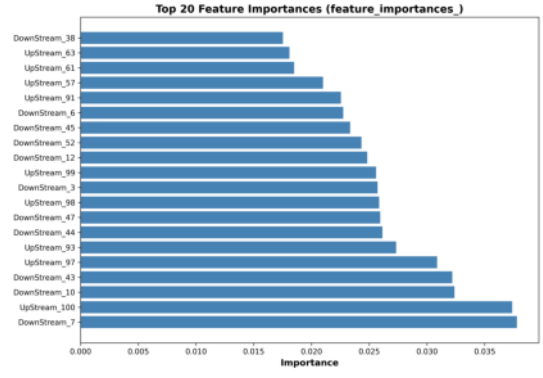
Performance Metrics

Accuracy	0.9333
Balanced Accuracy	0.9349
F1-Score (Macro)	0.9312
MSE	0.1667
MAE	0.1000
Log Loss	0.2802

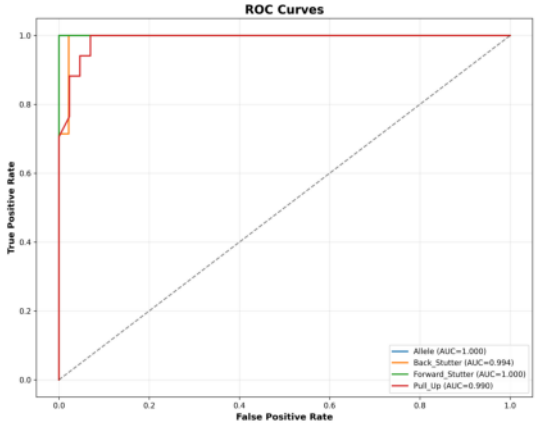
Confusion Matrix



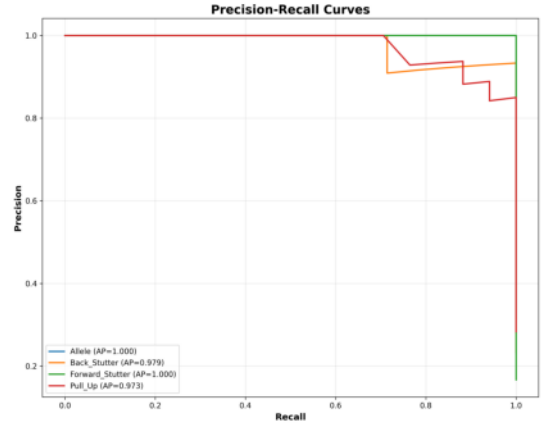
Feature Importance



ROC Curves

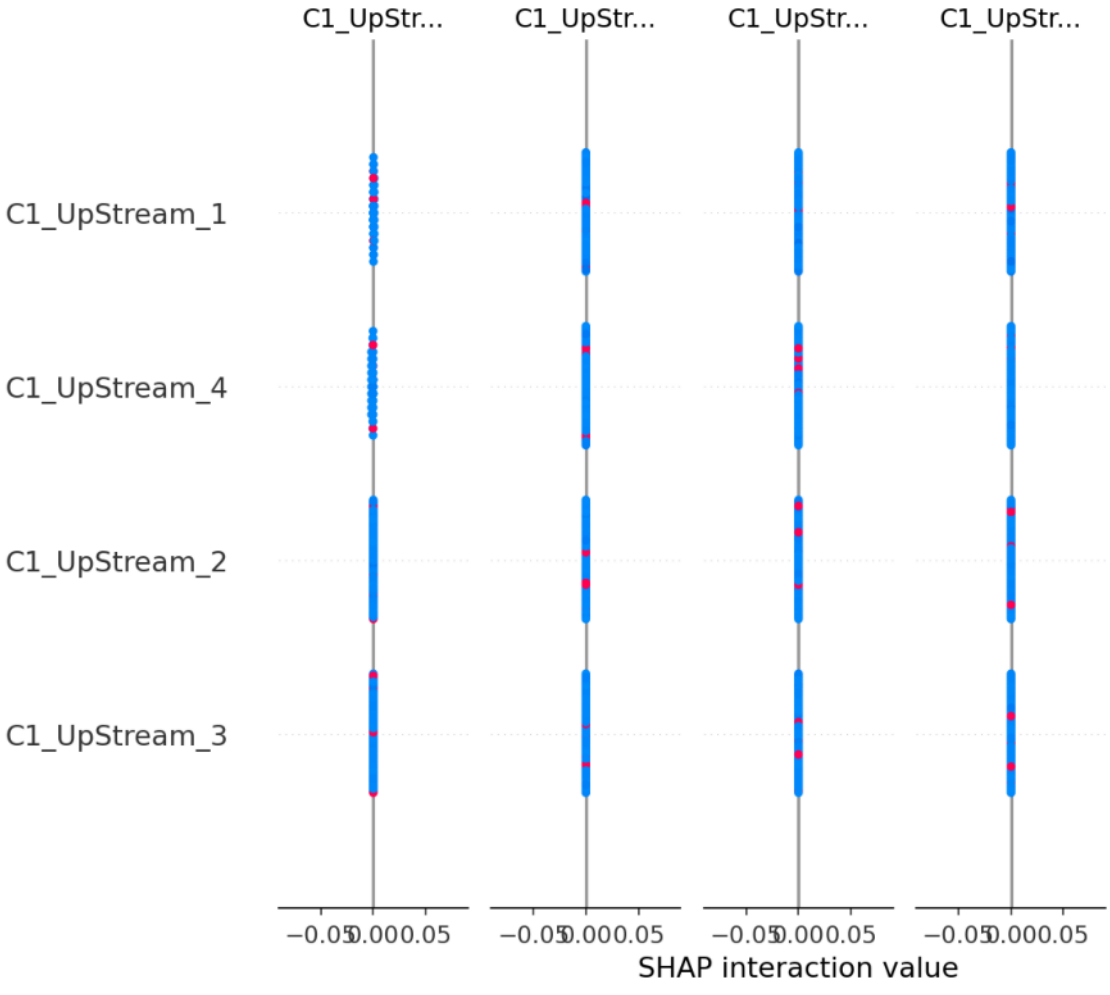


Precision-Recall Curves

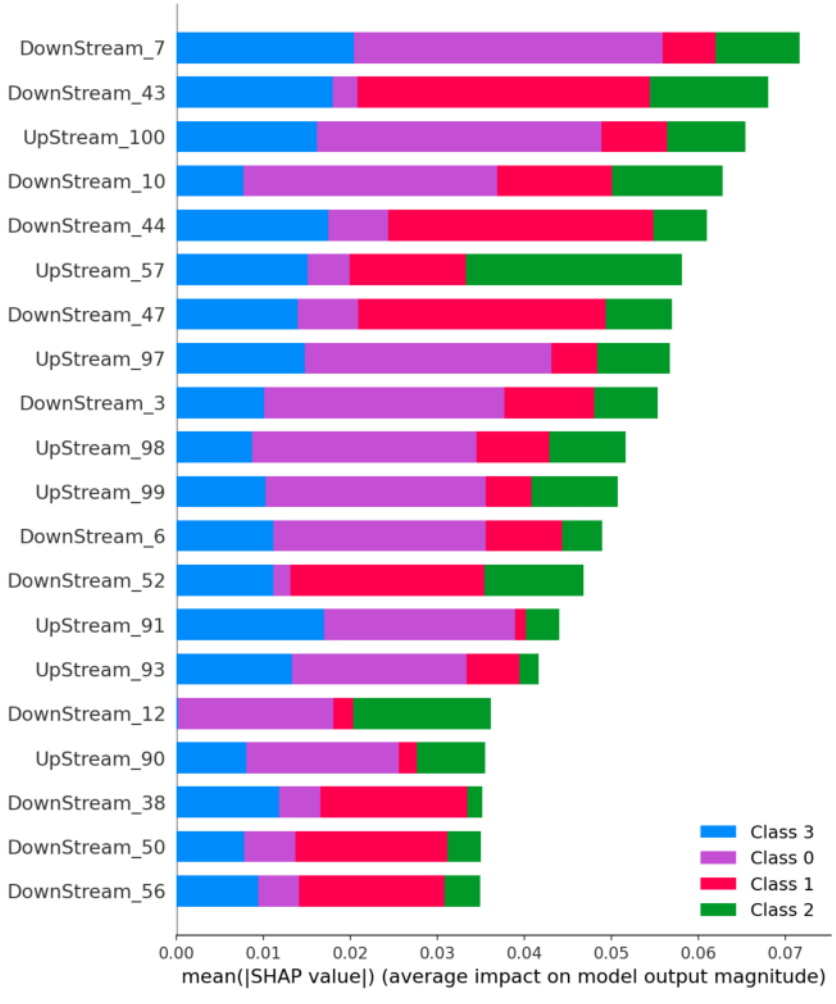


RANDOMFOREST - No Resampling - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

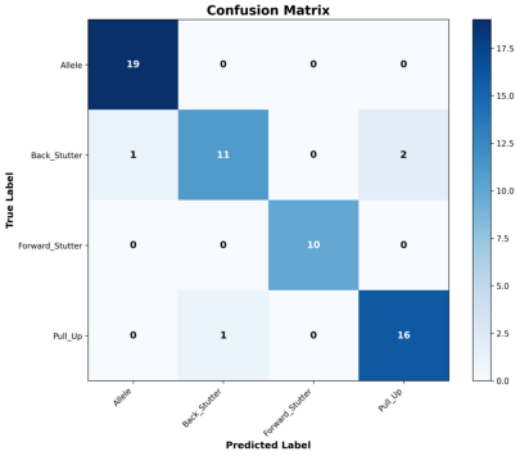


RANDOMFOREST - Class Weights Strategy

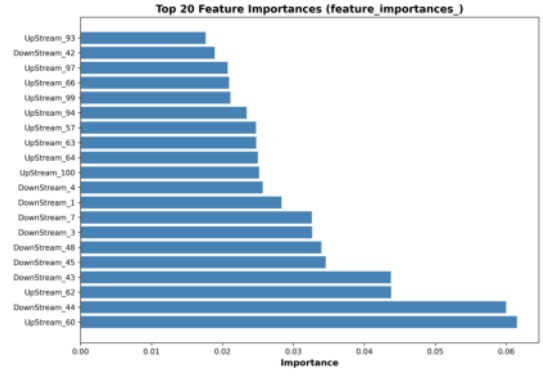
Performance Metrics

Accuracy	0.9333
Balanced Accuracy	0.9317
F1-Score (Macro)	0.9337
MSE	0.2167
MAE	0.1167
Log Loss	0.1850

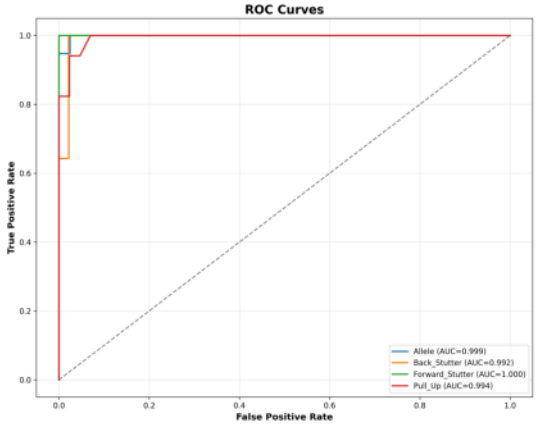
Confusion Matrix



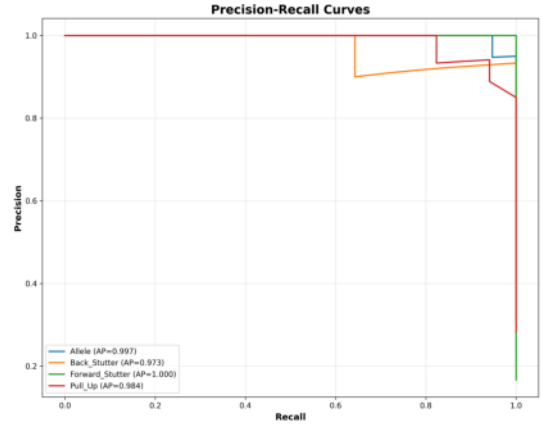
Feature Importance



ROC Curves

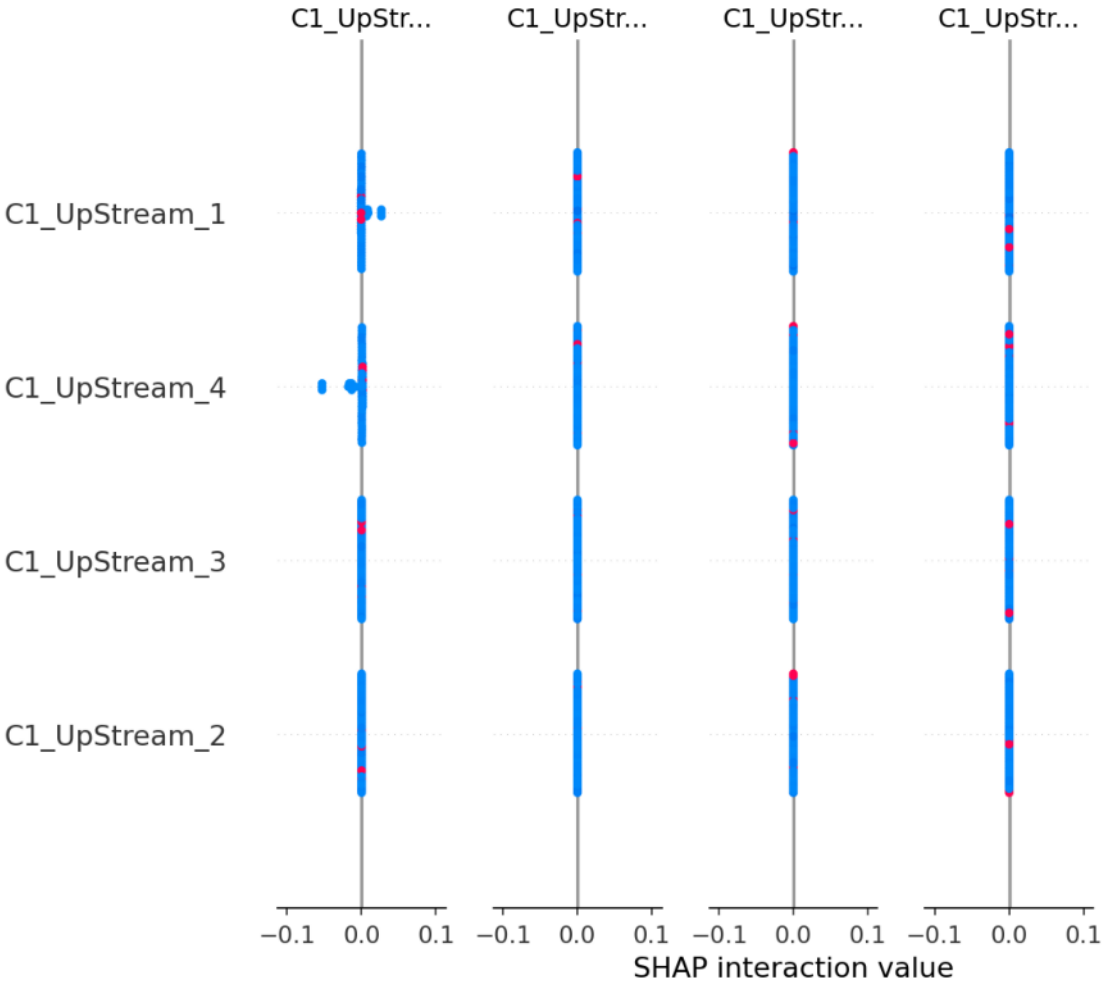


Precision-Recall Curves

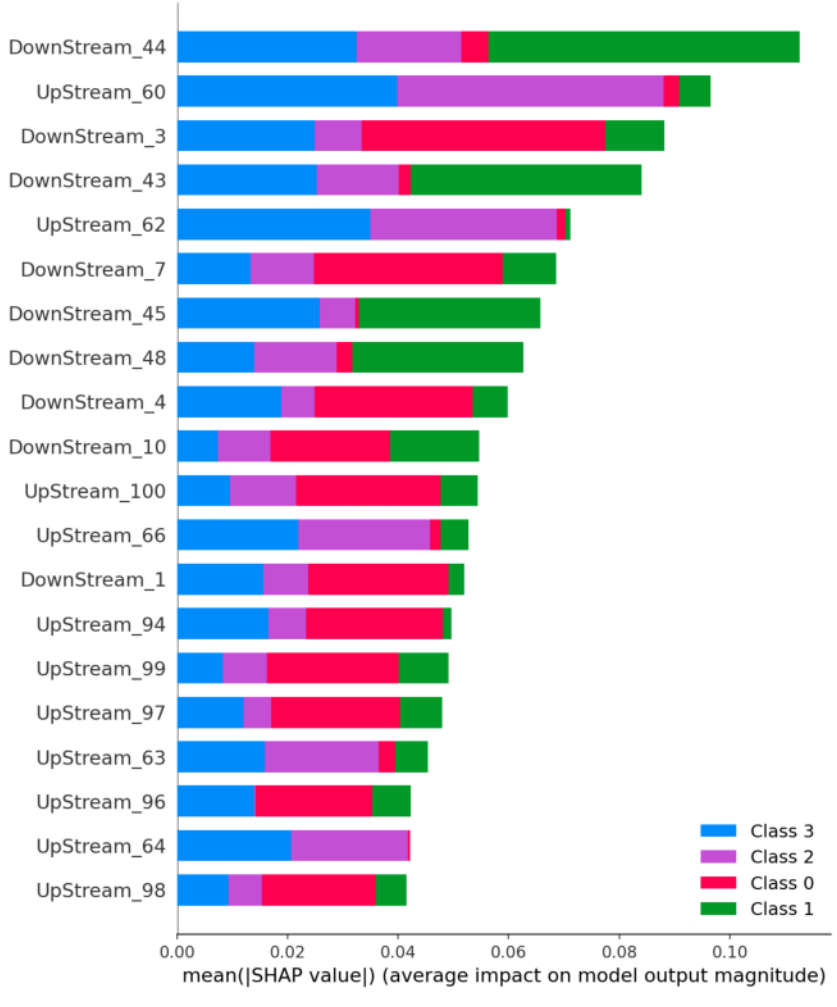


RANDOMFOREST - Class Weights - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

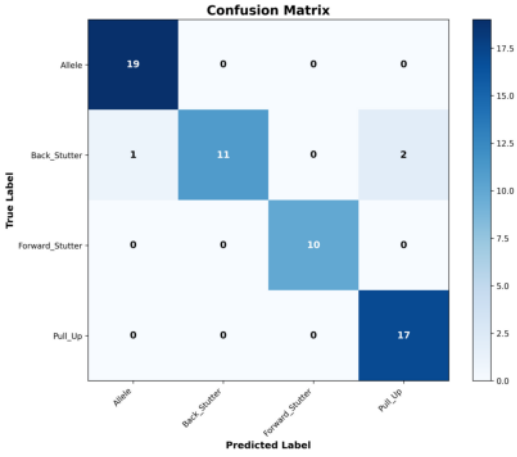


RANDOMFOREST - SMOTE Strategy

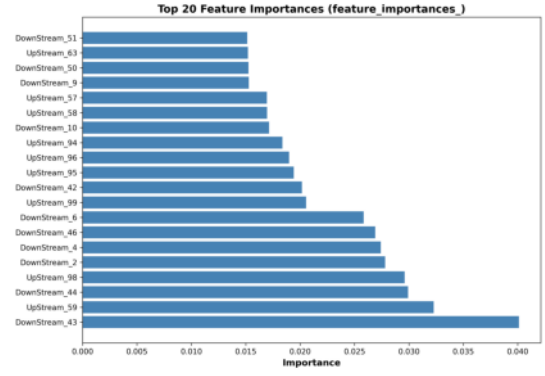
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9464
F1-Score (Macro)	0.9497
MSE	0.1500
MAE	0.0833
Log Loss	0.2101

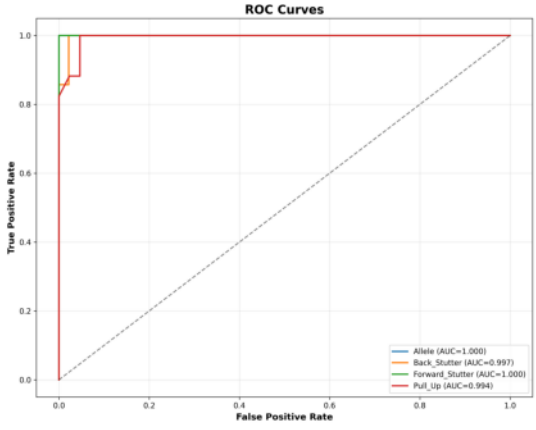
Confusion Matrix



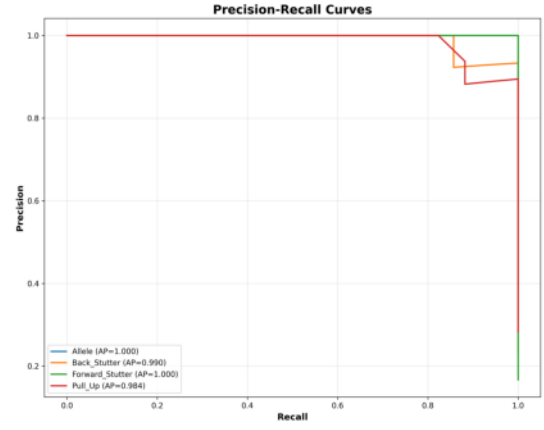
Feature Importance



ROC Curves

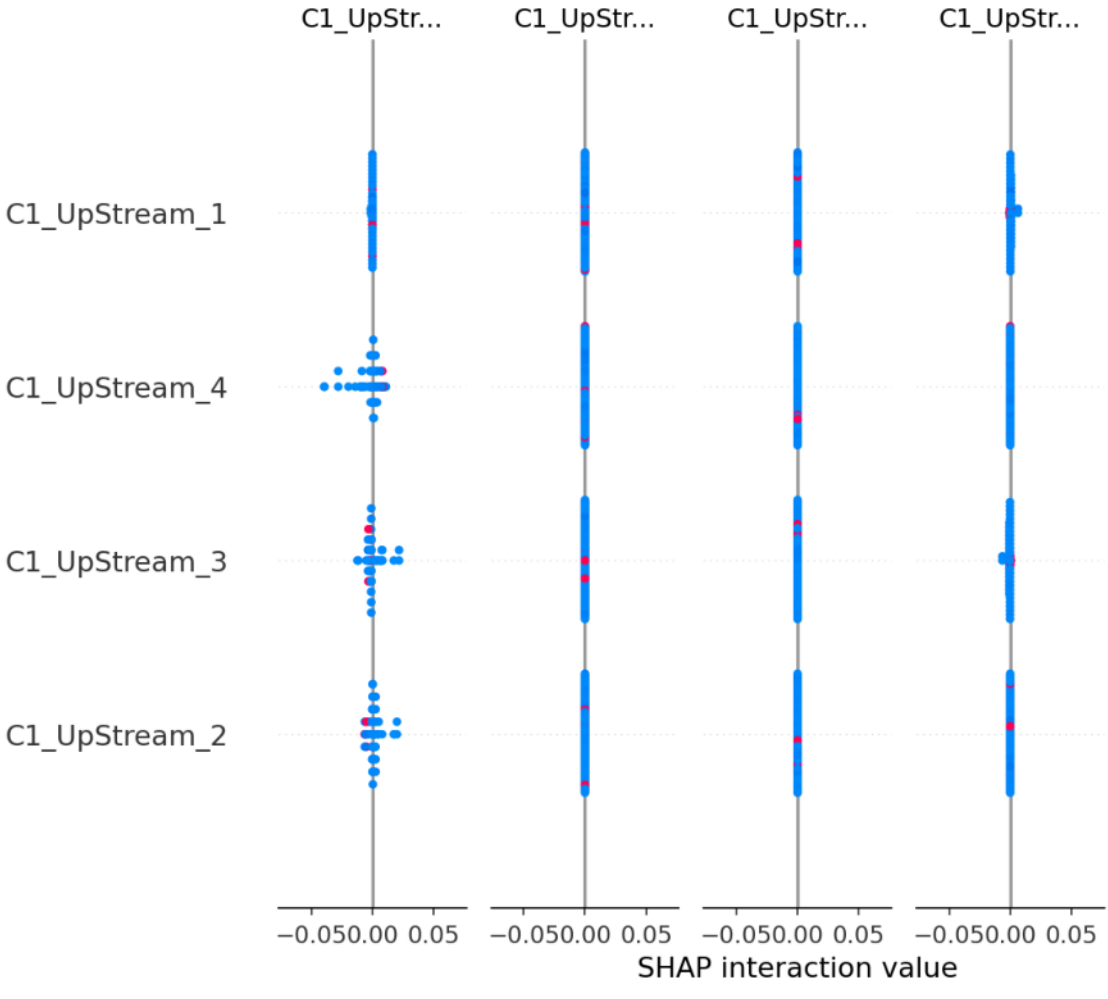


Precision-Recall Curves

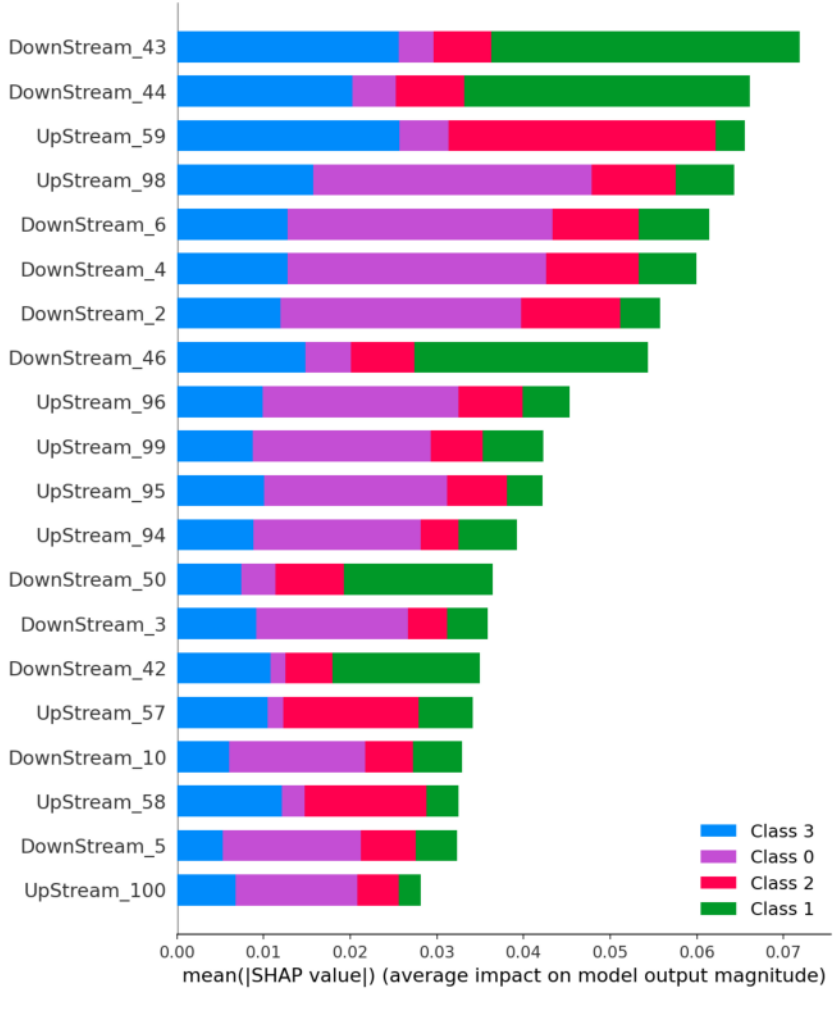


RANDOMFOREST - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

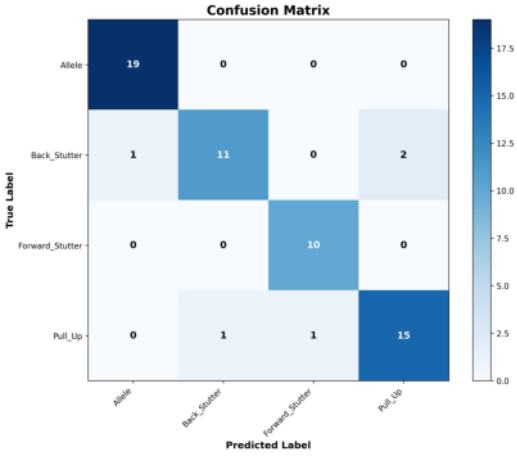


RANDOMFOREST - Borderline SMOTE Strategy

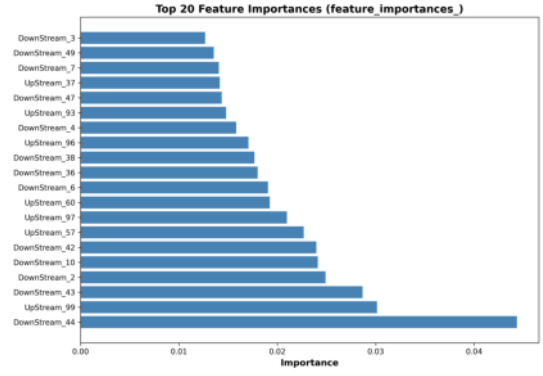
Performance Metrics

Accuracy	0.9167
Balanced Accuracy	0.9170
F1-Score (Macro)	0.9138
MSE	0.2333
MAE	0.1333
Log Loss	0.2913

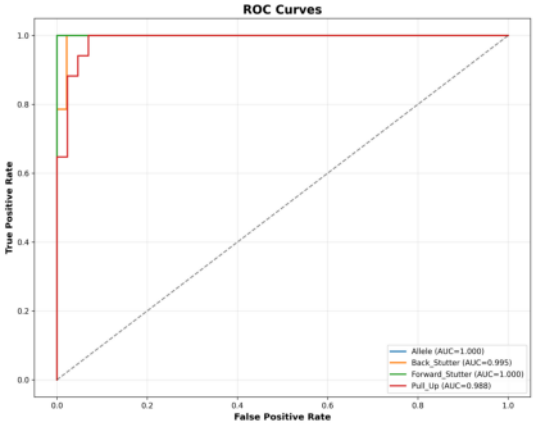
Confusion Matrix



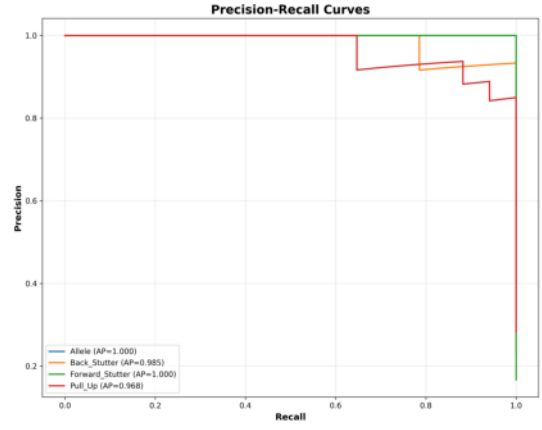
Feature Importance



ROC Curves

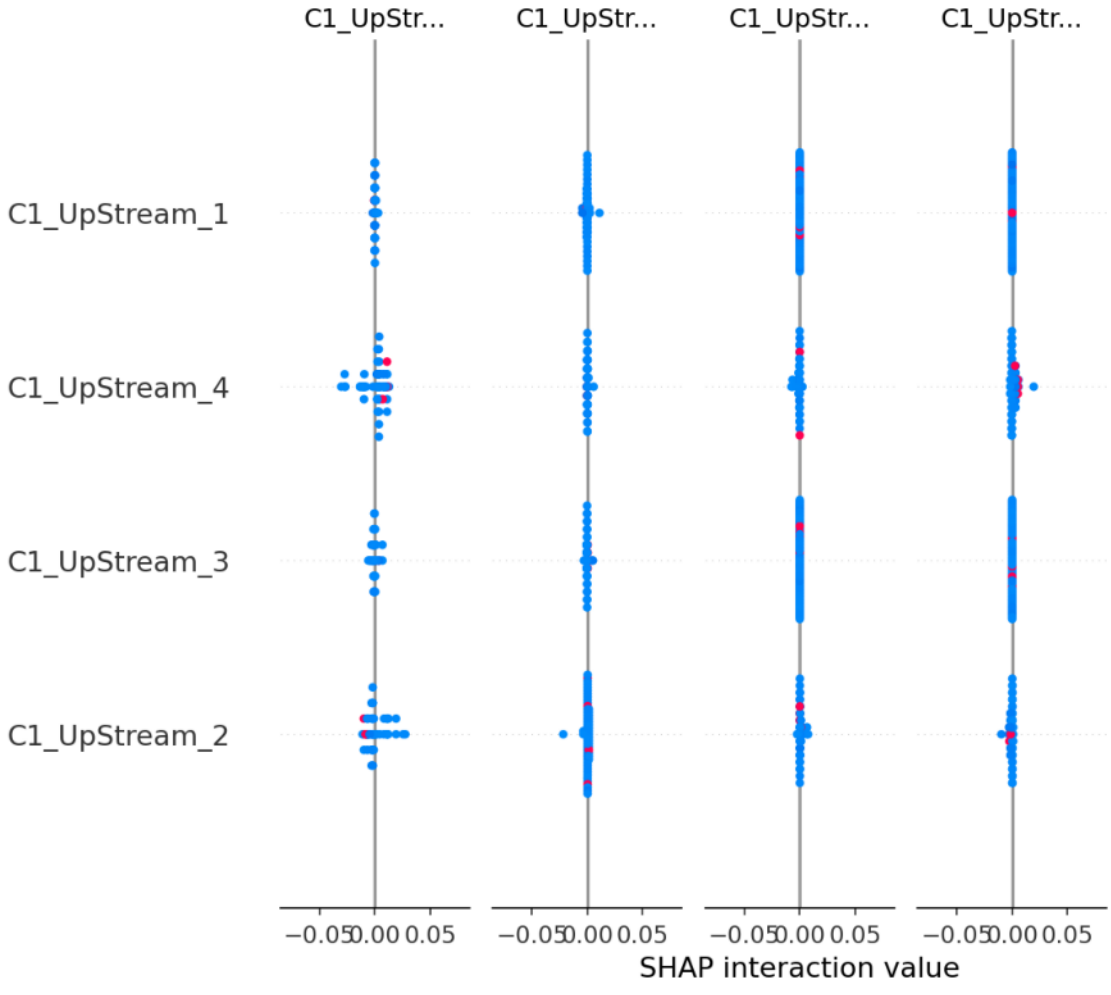


Precision-Recall Curves

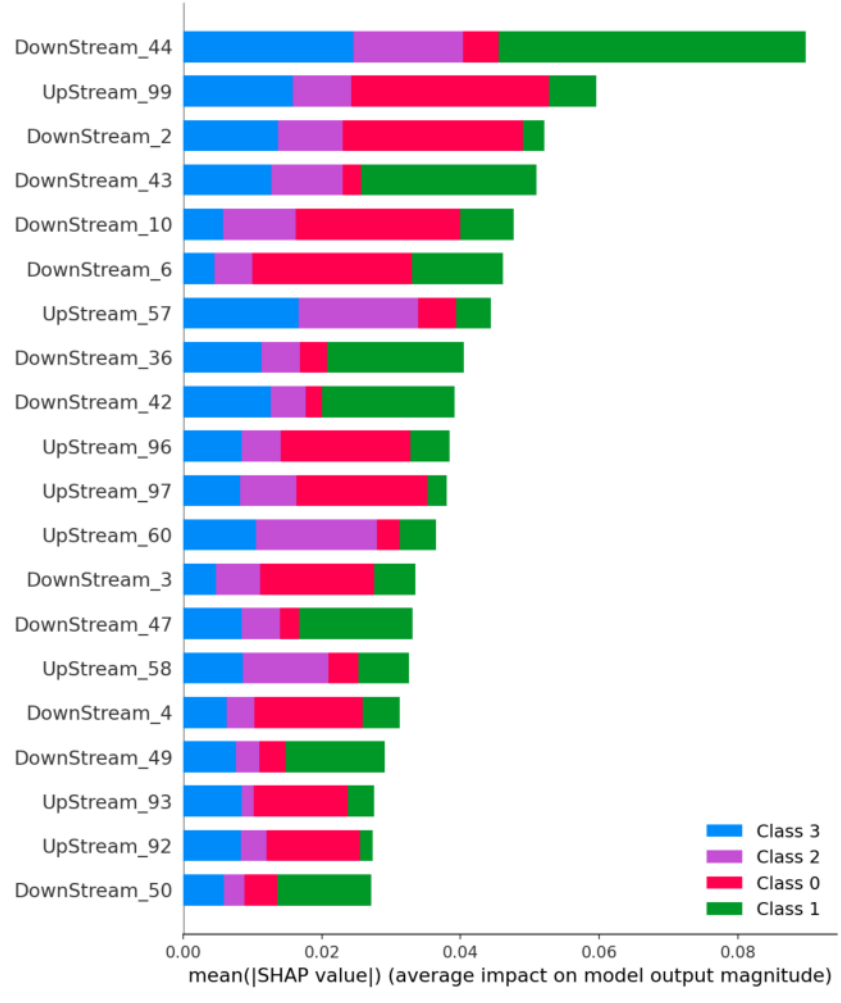


RANDOMFOREST - Borderline SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

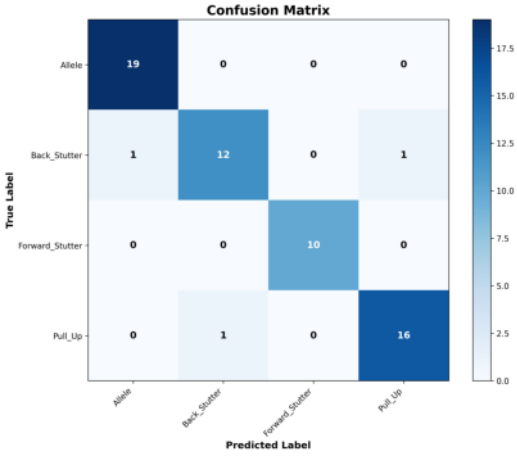


RANDOMFOREST - SVM SMOTE Strategy

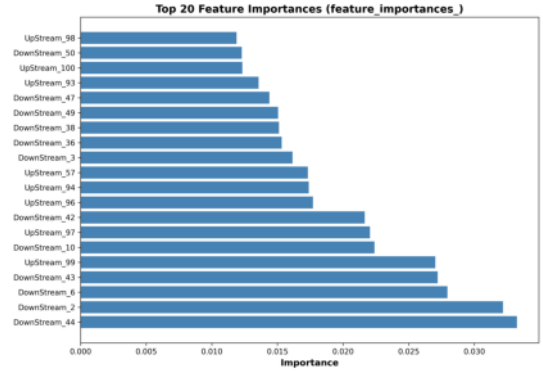
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9496
F1-Score (Macro)	0.9511
MSE	0.1500
MAE	0.0833
Log Loss	0.2414

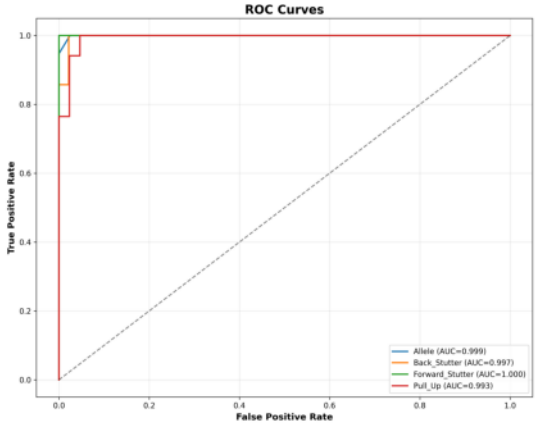
Confusion Matrix



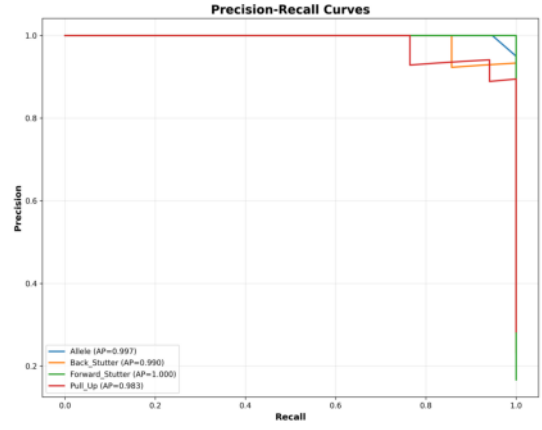
Feature Importance



ROC Curves

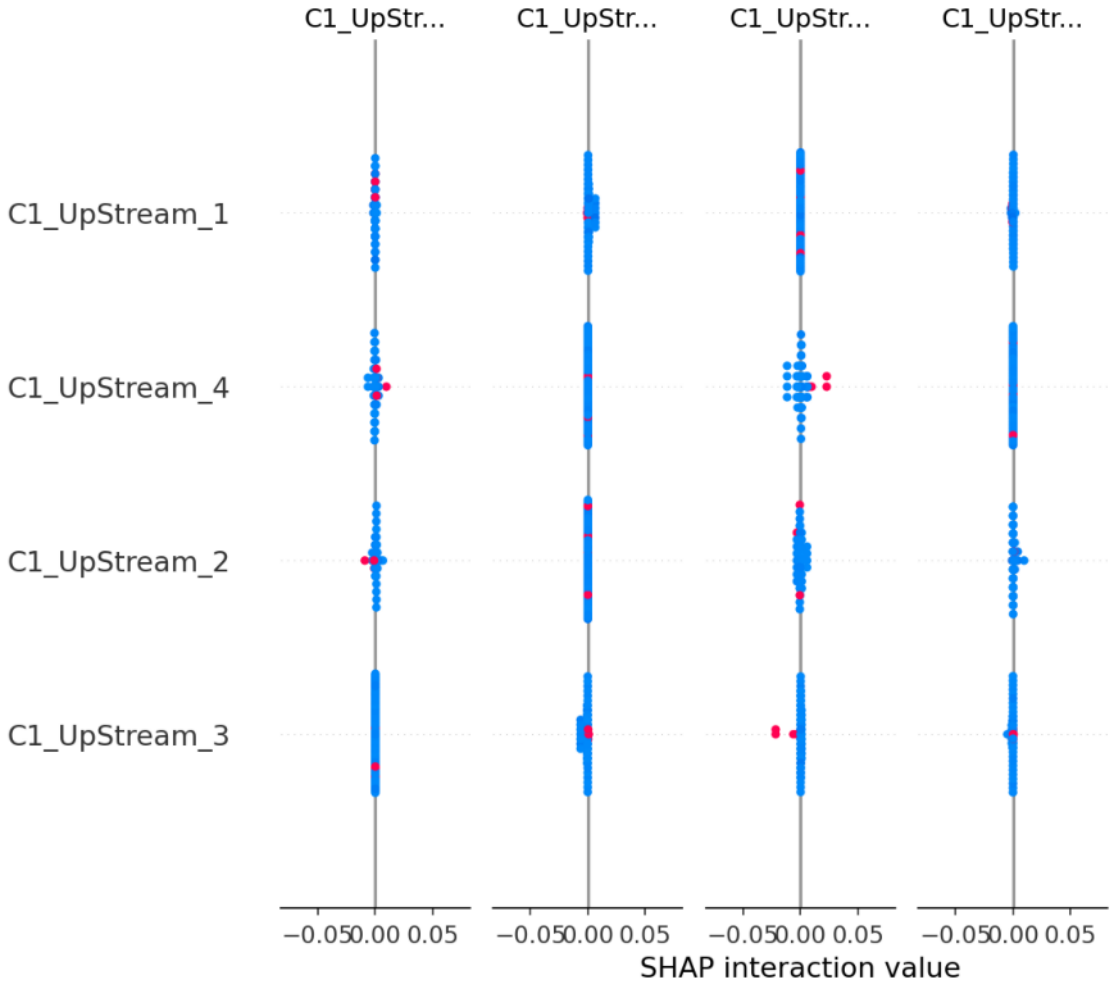


Precision-Recall Curves

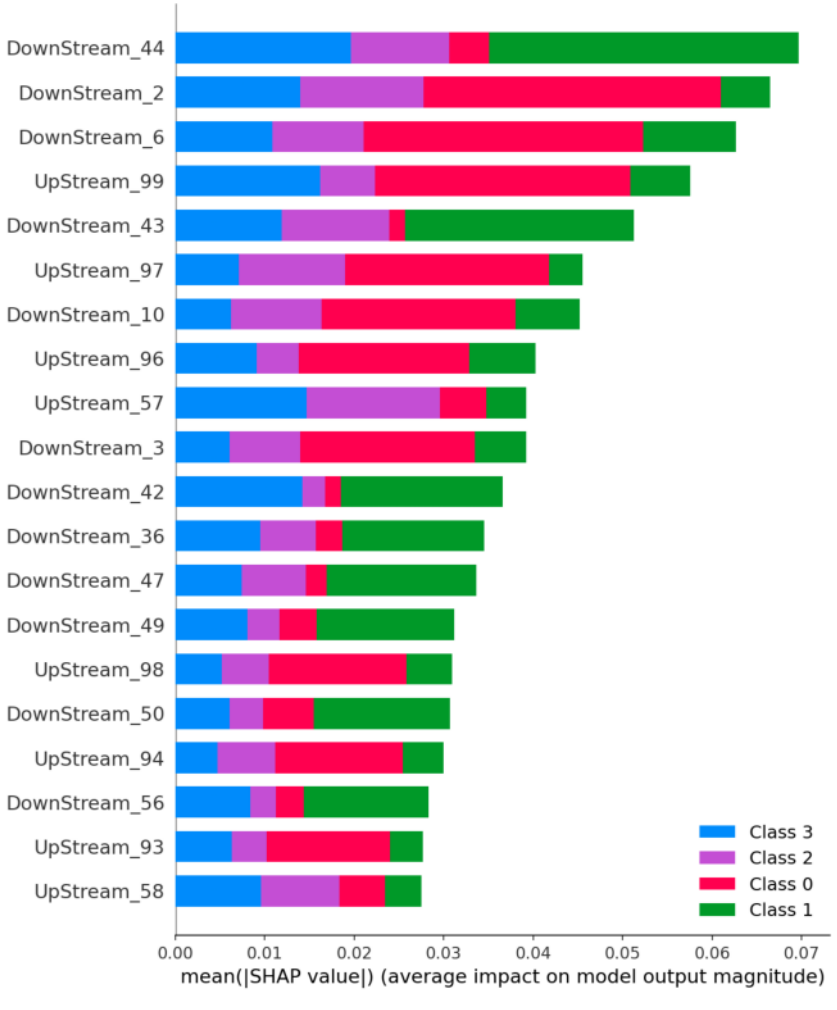


RANDOMFOREST - SVM SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

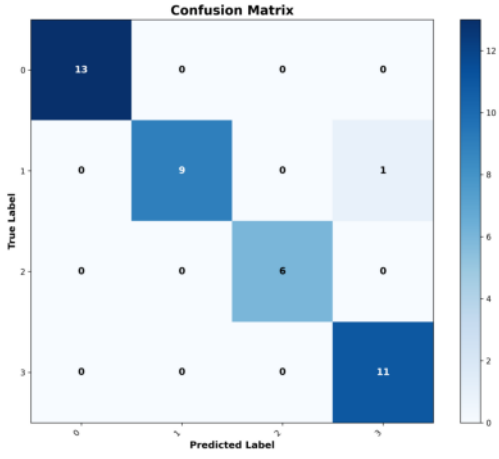


LOGISTIC_REGRESSION - Original Strategy

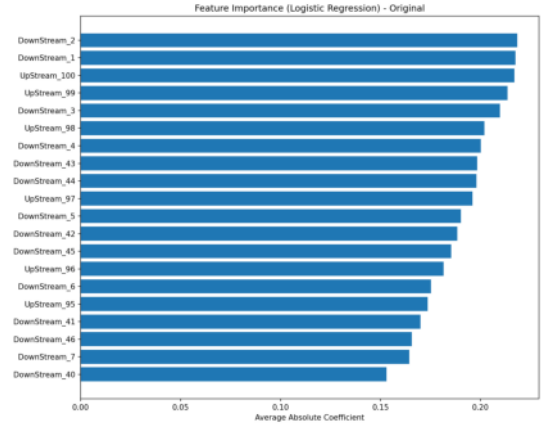
Performance Metrics

Accuracy	0.9750
Balanced Accuracy	0.9750
F1-Score (Macro)	0.9760
MSE	0.1000
MAE	0.0500
Log Loss	0.1021

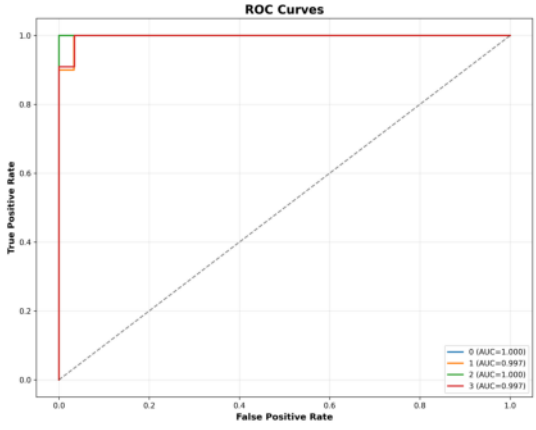
Confusion Matrix



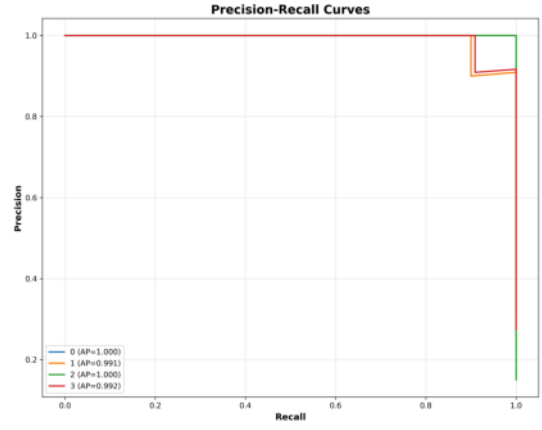
Feature Importance



ROC Curves

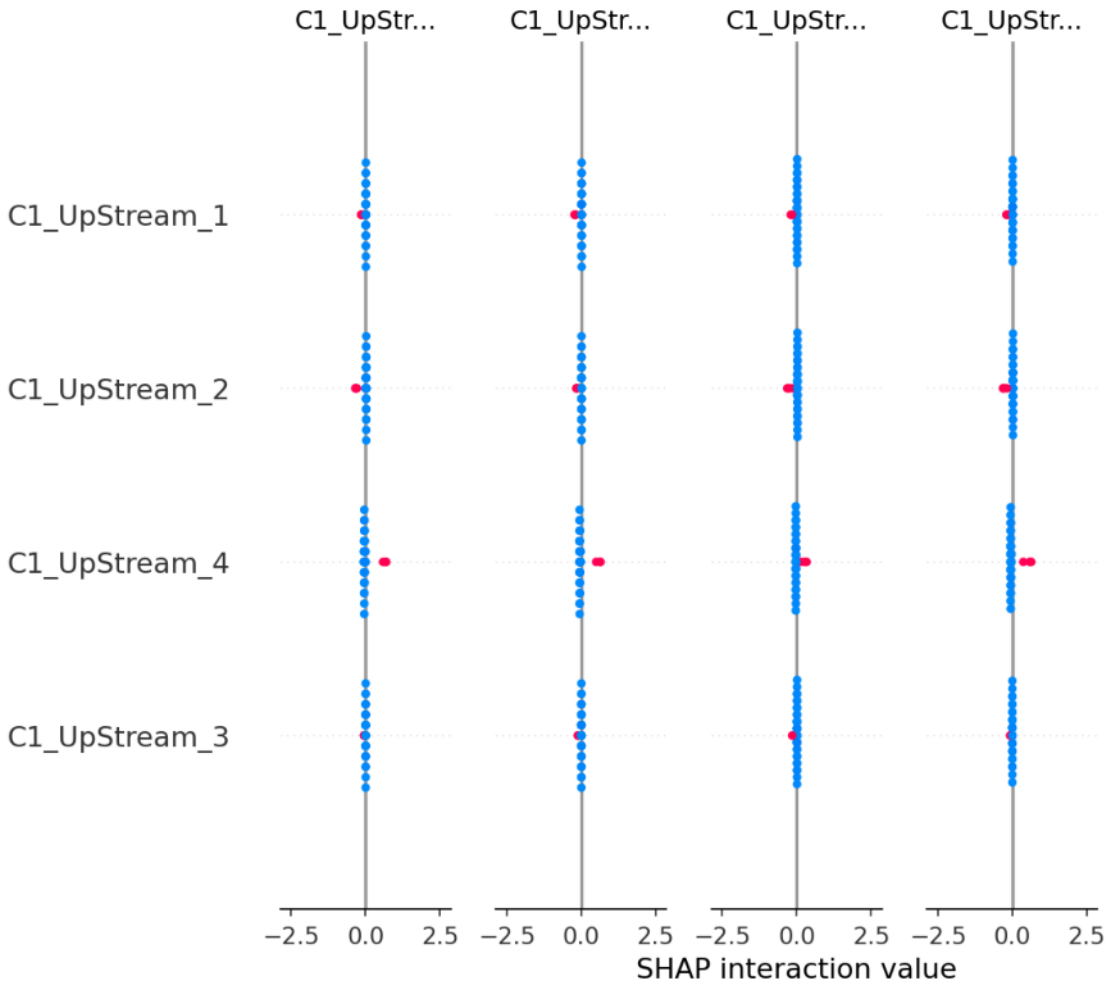


Precision-Recall Curves

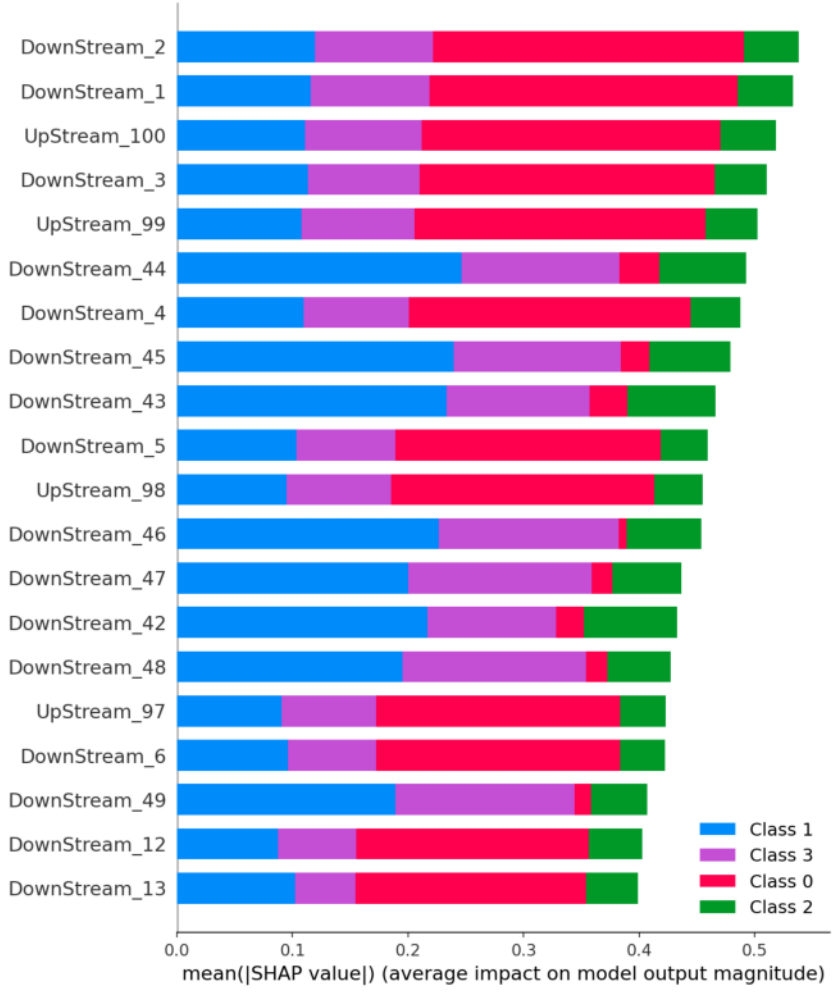


LOGISTIC_REGRESSION - Original - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

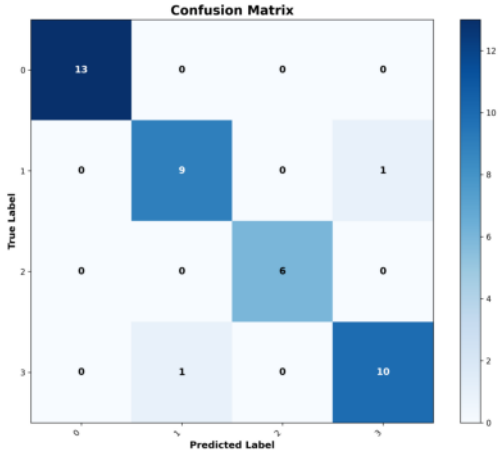


LOGISTIC_REGRESSION - SMOTE Strategy

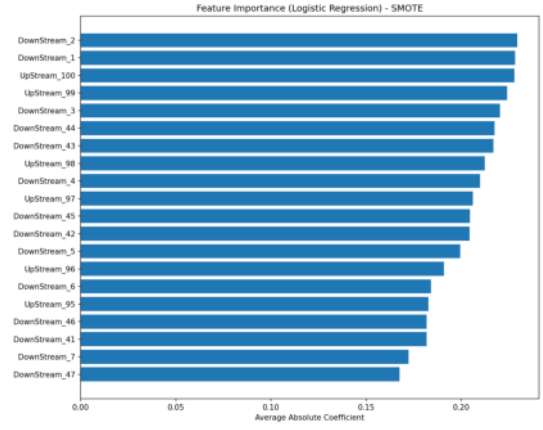
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9523
F1-Score (Macro)	0.9523
MSE	0.2000
MAE	0.1000
Log Loss	0.0941

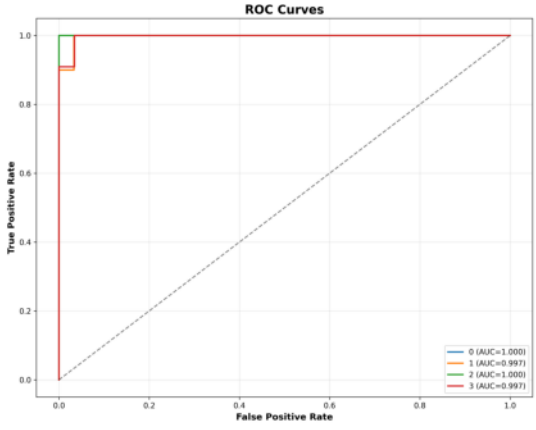
Confusion Matrix



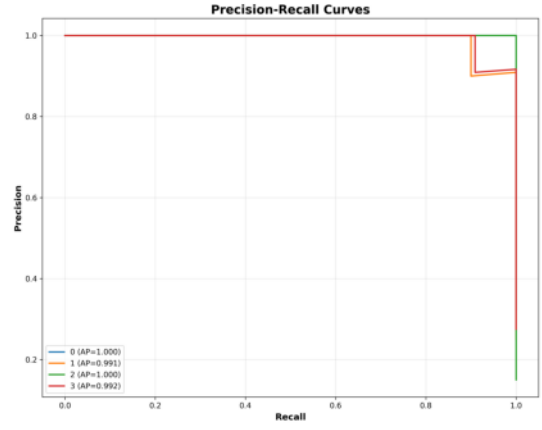
Feature Importance



ROC Curves

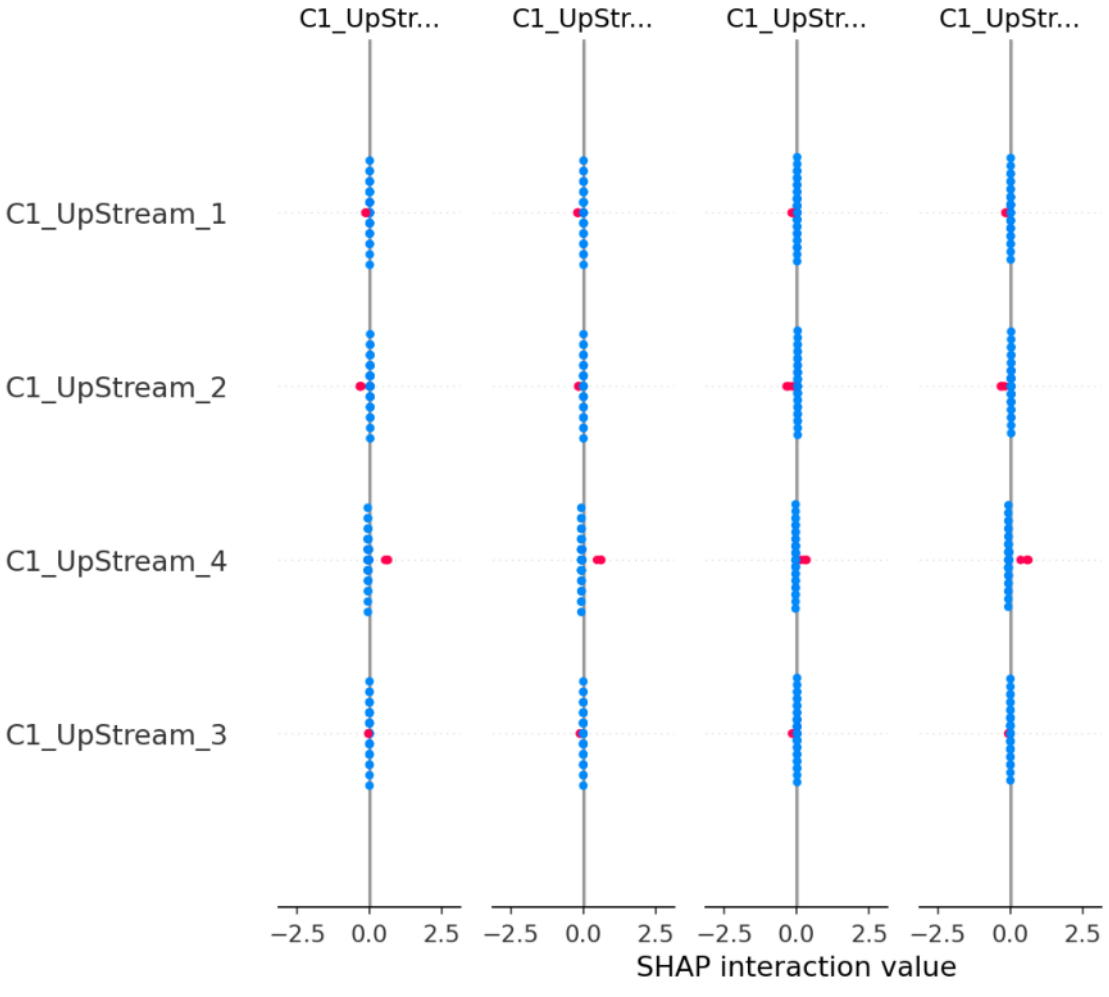


Precision-Recall Curves

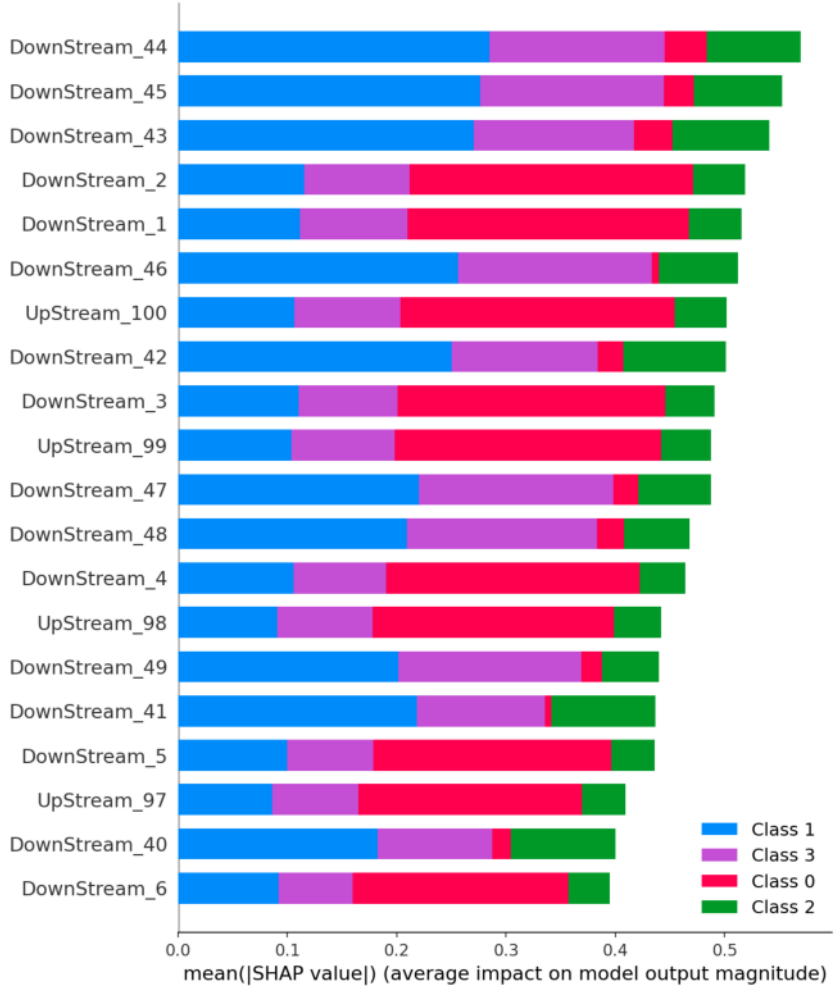


LOGISTIC_REGRESSION - SMOTE - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

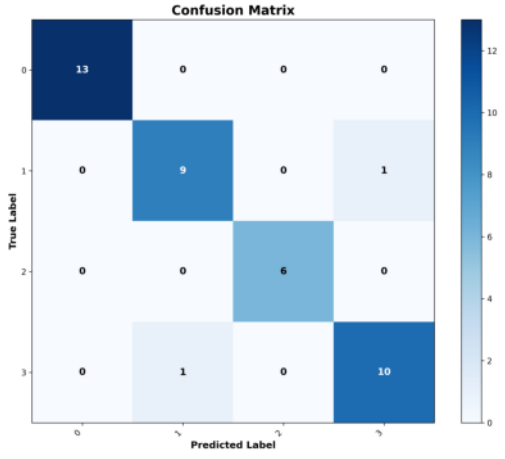


LOGISTIC_REGRESSION - RandomOverSampler Strategy

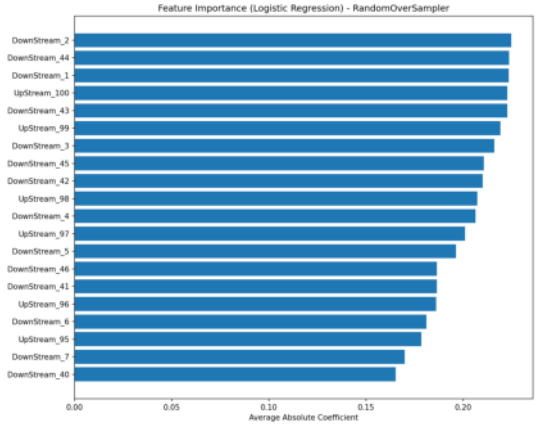
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9523
F1-Score (Macro)	0.9523
MSE	0.2000
MAE	0.1000
Log Loss	0.1193

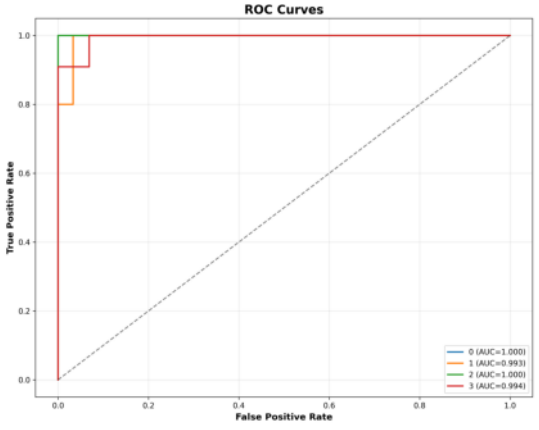
Confusion Matrix



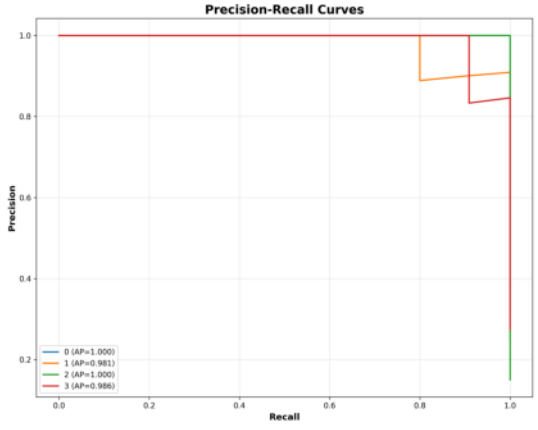
Feature Importance



ROC Curves

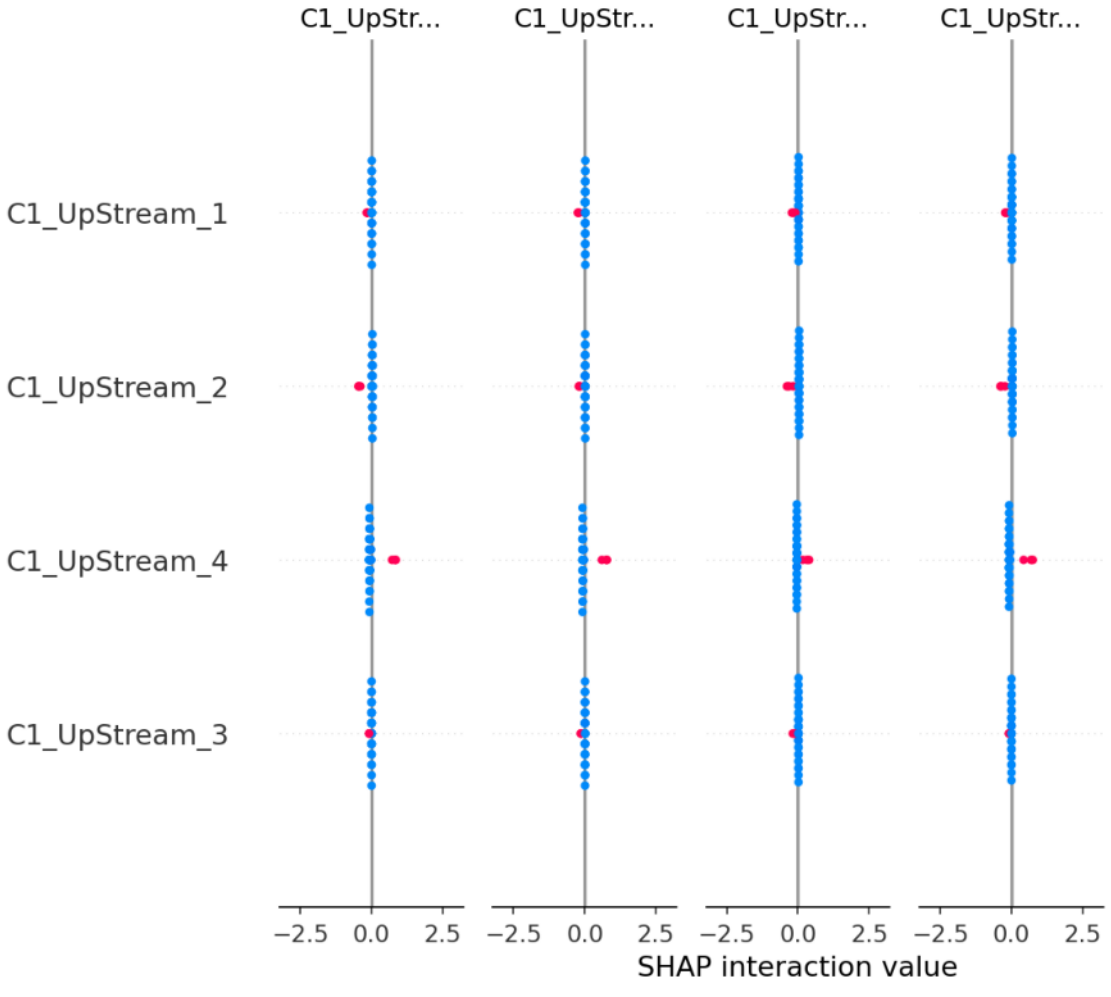


Precision-Recall Curves

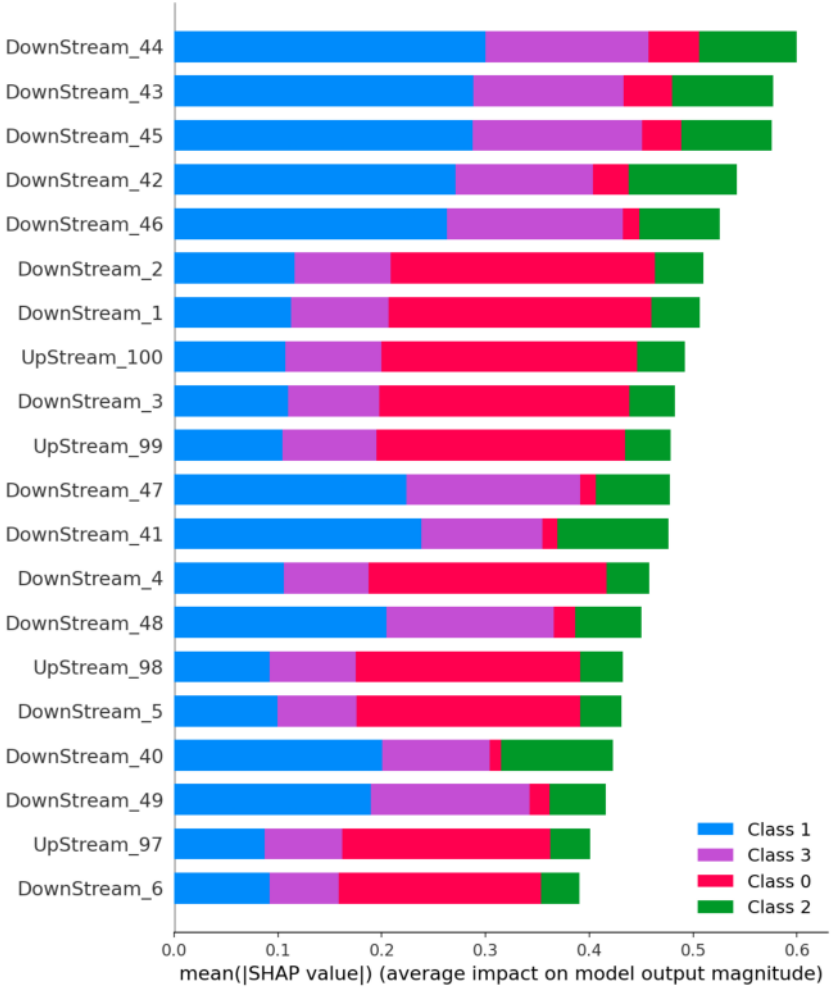


LOGISTIC_REGRESSION - RandomOverSampler - SHAP Analysis

SHAP Summary Plot



SHAP Bar Plot

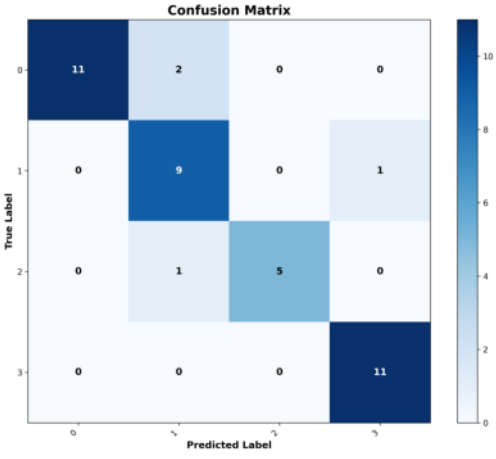


SVM - Original Strategy

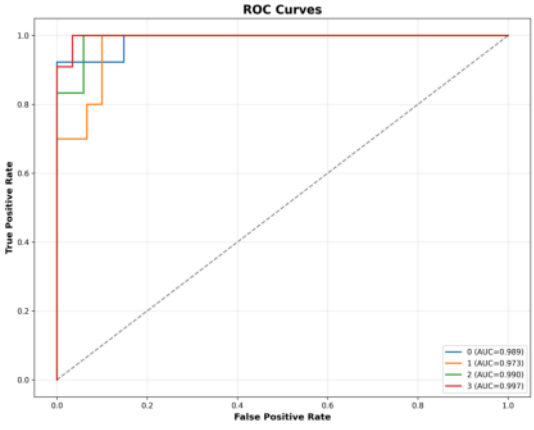
Performance Metrics

Accuracy	0.8250
Balanced Accuracy	0.8199
F1-Score (Macro)	0.8268
MSE	0.4750
MAE	0.2750
Log Loss	0.2955

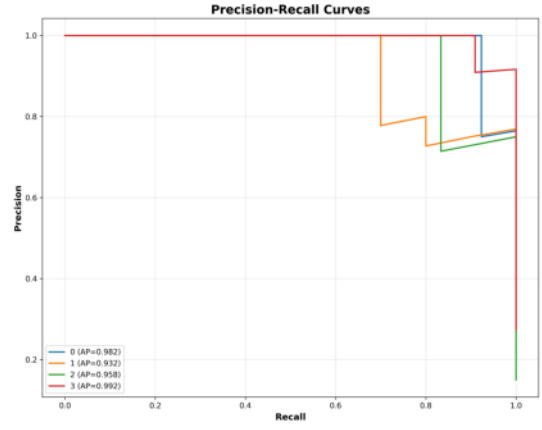
Confusion Matrix



ROC Curves



Precision-Recall Curves

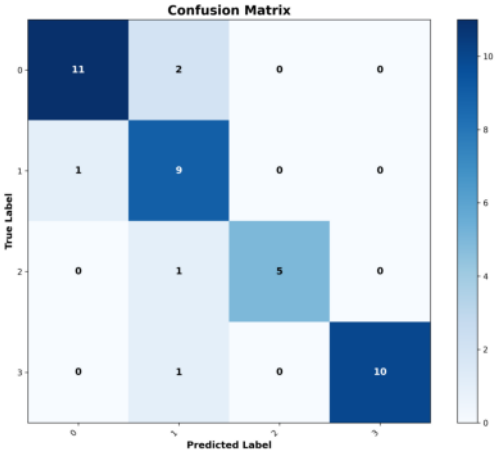


SVM - SMOTE Strategy

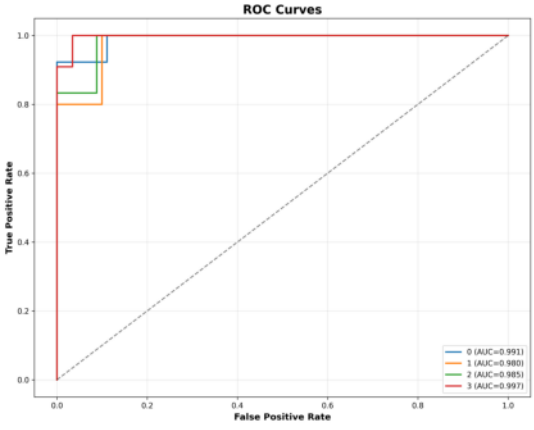
Performance Metrics

Accuracy	0.9500
Balanced Accuracy	0.9500
F1-Score (Macro)	0.9514
MSE	0.2000
MAE	0.1000
Log Loss	0.2764

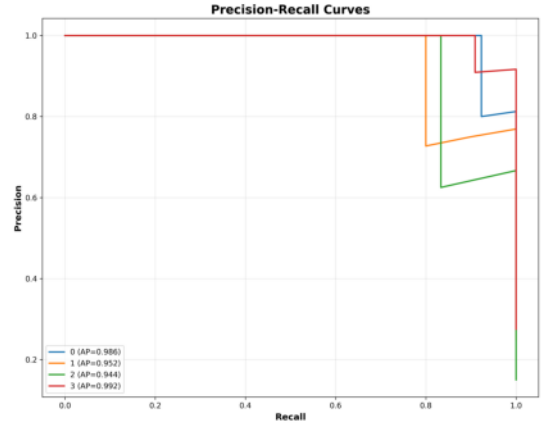
Confusion Matrix



ROC Curves



Precision-Recall Curves

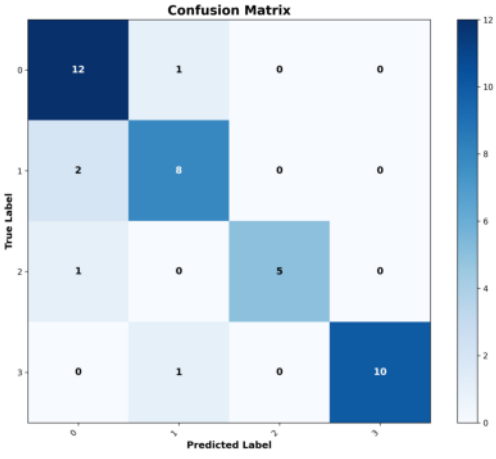


SVM - RandomOverSampler Strategy

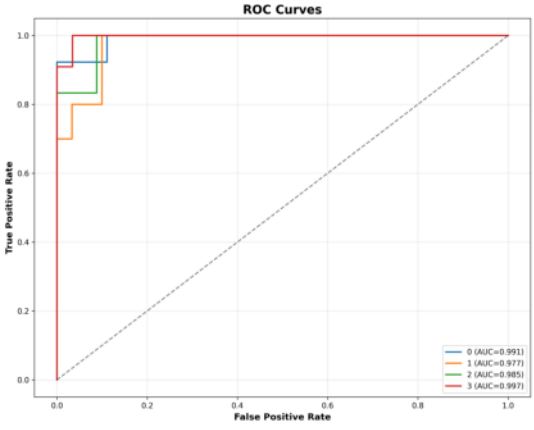
Performance Metrics

Accuracy	0.9250
Balanced Accuracy	0.9273
F1-Score (Macro)	0.9279
MSE	0.3000
MAE	0.1500
Log Loss	0.2732

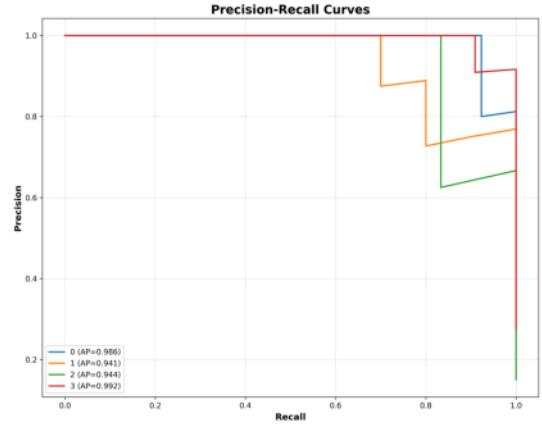
Confusion Matrix



ROC Curves



Precision-Recall Curves

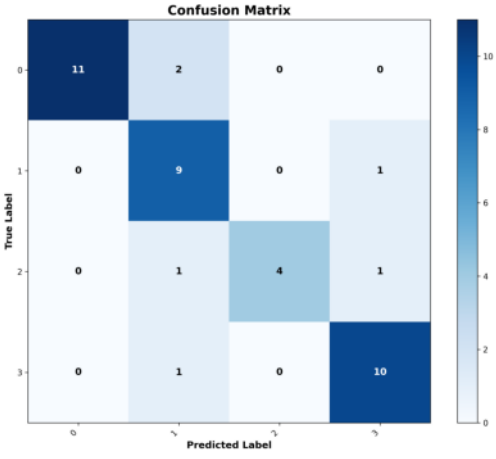


MLP - Original Strategy

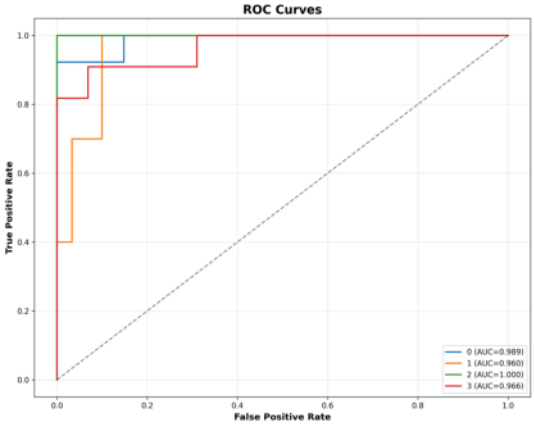
Performance Metrics

Accuracy	0.8250
Balanced Accuracy	0.8055
F1-Score (Macro)	0.8193
MSE	0.4000
MAE	0.2500
Log Loss	0.6329

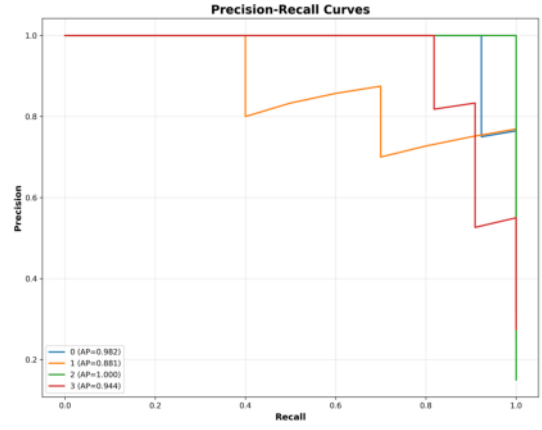
Confusion Matrix



ROC Curves



Precision-Recall Curves

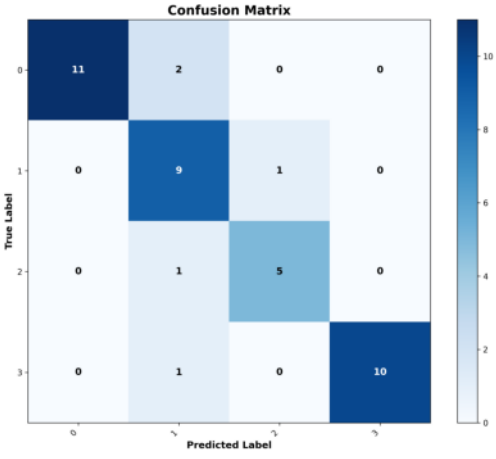


MLP - SMOTE Strategy

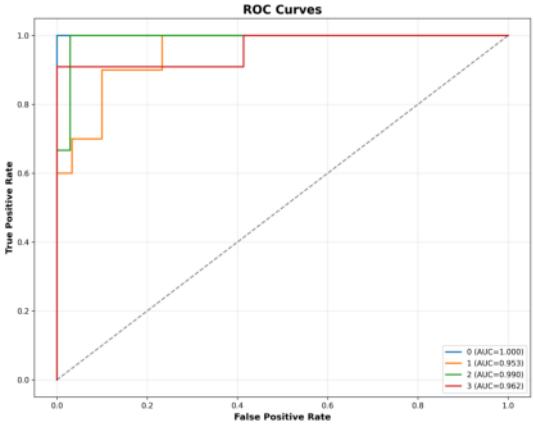
Performance Metrics

Accuracy	0.8500
Balanced Accuracy	0.8529
F1-Score (Macro)	0.8477
MSE	0.3750
MAE	0.2250
Log Loss	0.6846

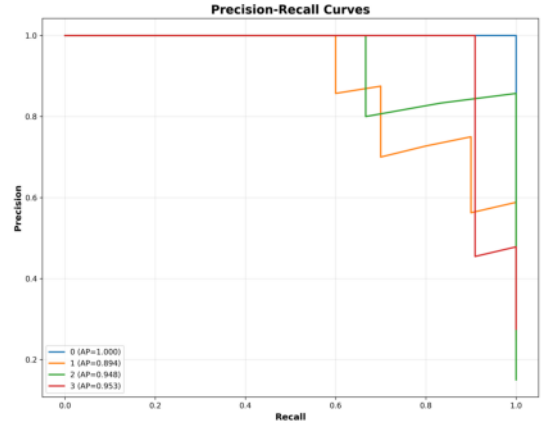
Confusion Matrix



ROC Curves



Precision-Recall Curves

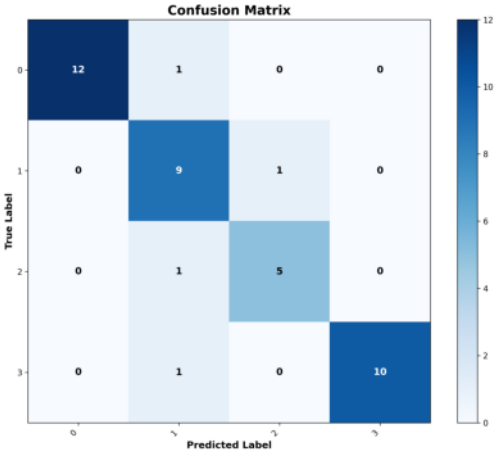


MLP - RandomOverSampler Strategy

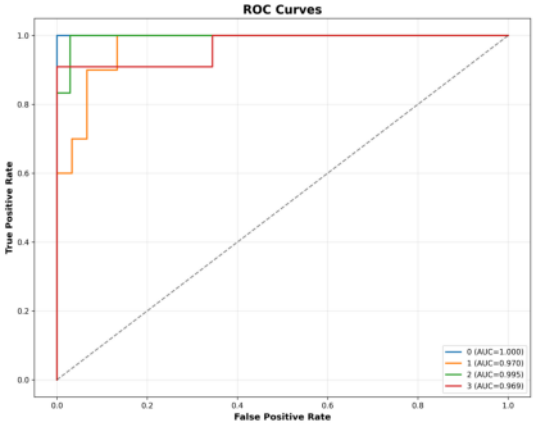
Performance Metrics

Accuracy	0.9250
Balanced Accuracy	0.9388
F1-Score (Macro)	0.9253
MSE	0.2250
MAE	0.1250
Log Loss	0.6314

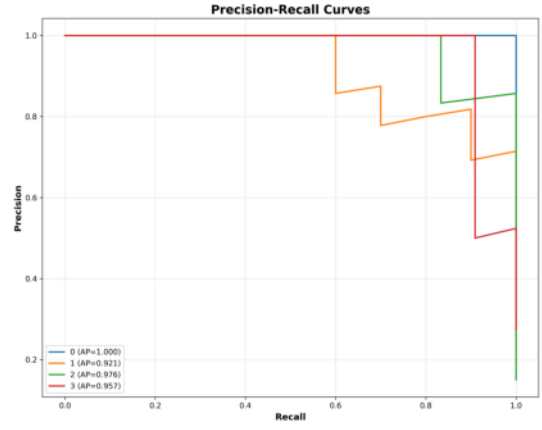
Confusion Matrix



ROC Curves



Precision-Recall Curves

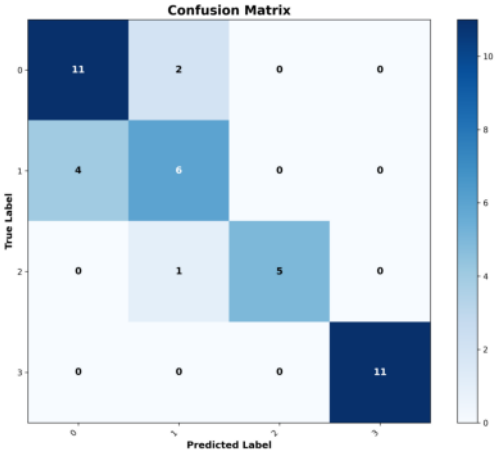


GNB - Original Strategy

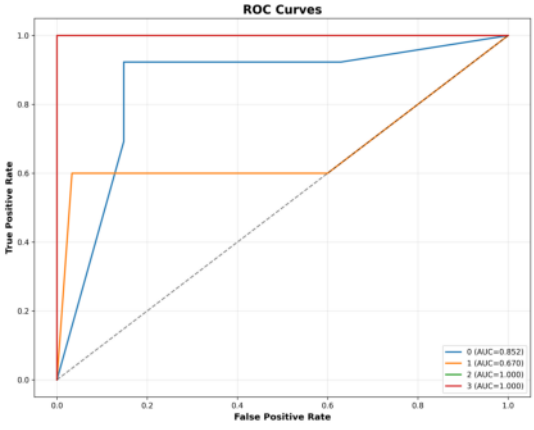
Performance Metrics

Accuracy	0.8250
Balanced Accuracy	0.8199
F1-Score (Macro)	0.8316
MSE	0.1750
MAE	0.1750
Log Loss	5.3056

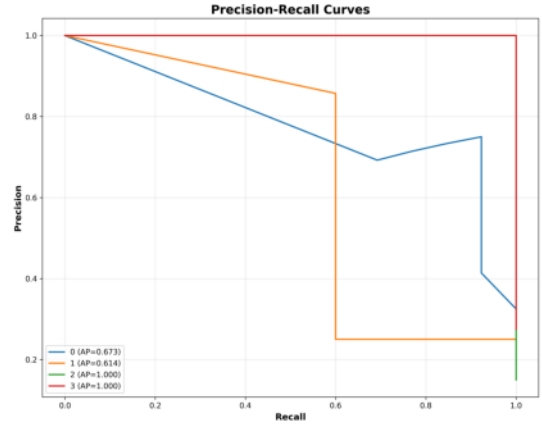
Confusion Matrix



ROC Curves



Precision-Recall Curves

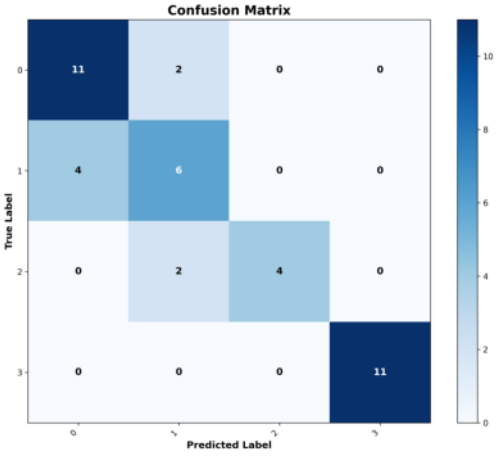


GNB - SMOTE Strategy

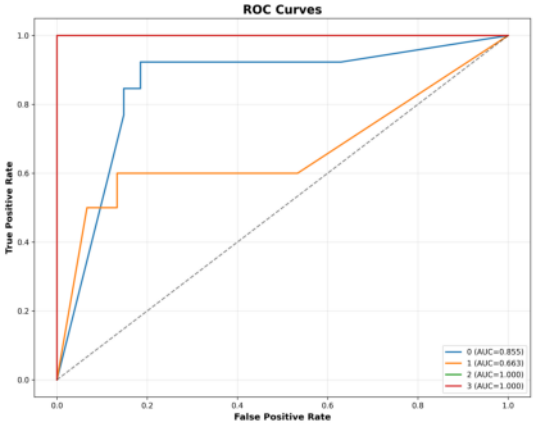
Performance Metrics

Accuracy	0.8000
Balanced Accuracy	0.7782
F1-Score (Macro)	0.7964
MSE	0.2000
MAE	0.2000
Log Loss	6.4329

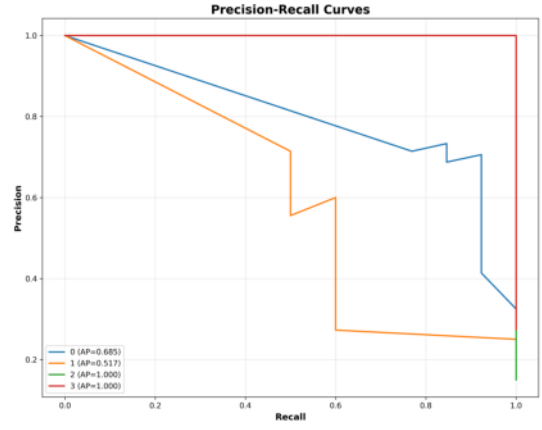
Confusion Matrix



ROC Curves



Precision-Recall Curves

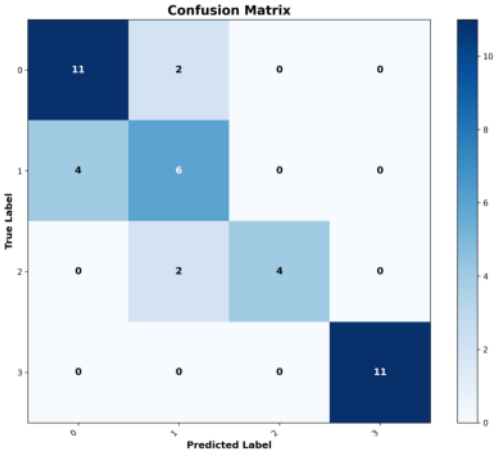


GNB - RandomOverSampler Strategy

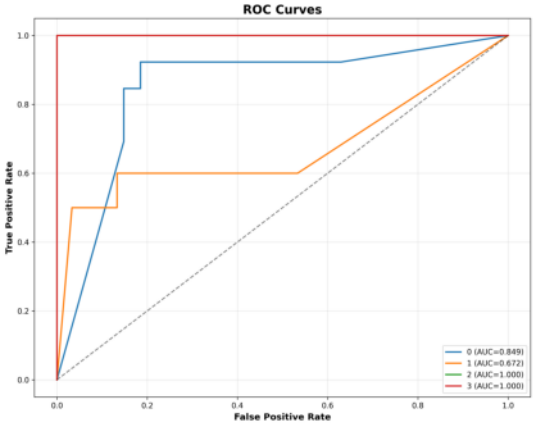
Performance Metrics

Accuracy	0.8000
Balanced Accuracy	0.7782
F1-Score (Macro)	0.7964
MSE	0.2000
MAE	0.2000
Log Loss	5.8224

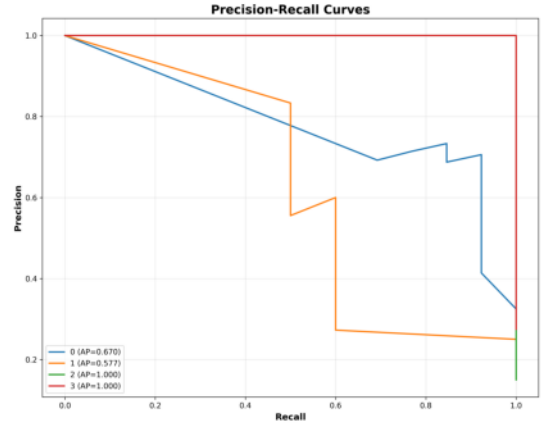
Confusion Matrix



ROC Curves



Precision-Recall Curves



Conclusions and Recommendations

KEY FINDINGS

Best Performing Model: TABPFN with Original strategy
• Balanced Accuracy: 0.9750
• F1-Score (Macro): 0.9760
• Accuracy: 0.9750

OBSERVATIONS

1. RESAMPLING IMPACT

Different models respond differently to resampling strategies. SMOTE and RandomOverSampler can help with class imbalance, but may lead to overfitting in some cases.

2. MODEL COMPLEXITY

Tree-based models (Random Forest, XGBoost, LightGBM, Extra Trees) generally perform well on this dataset and provide interpretability through feature importance and SHAP values.

3. CLASS IMBALANCE

The dataset shows class imbalance. Balanced accuracy and per-class metrics provide better insights than overall accuracy alone.

4. FEATURE IMPORTANCE

SHAP values and feature importance plots reveal which features most strongly influence DNA artifact classification. Review these for domain validation.

RECOMMENDATIONS

✓ Use the best performing model for production deployment
✓ Monitor per-class performance, especially for minority classes
✓ Consider ensemble methods combining top performers
✓ Validate feature importance with domain experts
✓ Collect more data for underrepresented classes if possible
✓ Perform cross-validation for more robust performance estimates
✓ Test on independent validation set before deployment

NEXT STEPS

→ Hyperparameter tuning for top models
→ Cross-validation analysis
→ External validation dataset testing
→ Error analysis for misclassified samples
→ Feature engineering based on SHAP insights
→ Consider model calibration for probability estimates